

Honeywell

Experion

**Planning, Installation, and Service
for WS360/WS370/WS470**

EX14-100

EP-DPCXX1

Release Independent

07/05

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About This Document

This document contains installation and service information for the WS360, WS370, and WS470 computer platforms.

Release Information

Document Name	Document ID	Release Number	Publication Date
Planning, Installation, and Service for WS360/WS370/WS470 – EX14	EP-DPCXX1	Release Independent	07/05

References

The following list identifies all documents that may be sources of reference for material discussed in this publication.

Document Title
TPS Users
<i>TPS System Site Planning</i>
<i>TPS System Implementation Guide for Windows 2000</i>
<i>TPS System Planning Guide for Windows 2000</i>
<i>TPS System Administration Guide for Windows 2000</i>
Experion PKS Users
<i>Experion PKS Overview</i>
<i>Experion PKS Software Installation and Upgrade Guide</i>
<i>Server and Client Planning Guide</i>
<i>Server and Client Configuration Guide (for Experion PKS)</i>
<i>Experion PKS Operators Guide</i>
FTE Users
<i>Fault Tolerant Ethernet Installation and Service Guide e</i>
Console Documents
<i>Honeywell Icon Series Console Planning, Installation, and Service Guide</i>
<i>Universal Station Service</i>
<i>Universal Station (Ergonomic) Service</i>

Contacts

World Wide Web

The following lists Honeywell's World Wide Web sites that will be of interest to our Process Solutions customers.

Honeywell Organization	WWW Address (URL)
Corporate	http://www.honeywell.com
Process Solutions	http://www.acs.honeywell.com
International	http://content.honeywell.com/global/

Telephone

Contact us by telephone at the numbers listed below.

Organization		Phone Number	
United States and Canada	Honeywell International Inc. Process Solutions	1-800-223-8947	Sales
		1-800-525-7439	Service
		1-800-822-7673	Tech. Support
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Europe	Honeywell PACE Brussels, Belgium	[32-2] 728-2711	
Latin America	Honeywell International Inc. Sunrise, Florida U.S.A.	(954) 845-2600	

Symbol Definitions

The following table lists those symbols used in this document to denote certain conditions.

Symbol	Definition
	ATTENTION: Identifies information that requires special consideration.
	TIP: Identifies advice or hints for the user, often in terms of performing a task.
	REFERENCE -EXTERNAL: Identifies an additional source of information outside of the book set.
	REFERENCE - INTERNAL: Identifies an additional source of information within the book set.
CAUTION	Indicates a situation which, if not avoided, may result in equipment or work (data) on the system being damaged or lost, or may result in the inability to properly operate the process.
	CAUTION: Indicates a potentially hazardous situation, which if not avoided, may result in minor or moderate injury. It may also be used to alert against unsafe practices. CAUTION symbol on the equipment refers the user to the product manual for additional information. The symbol appears next to required information in the manual.
	WARNING: Indicates a potentially hazardous situation, which if not avoided, could result in serious injury or death. WARNING symbol on the equipment refers the user to the product manual for additional information. The symbol appears next to required information in the manual.
	WARNING, Risk of electrical shock: Potential shock hazard where HAZARDOUS LIVE voltages greater than 30 Vrms, 42.4 Vpeak, or 60 VDC may be accessible.

Symbol	Definition
	ESD HAZARD: Danger of an electro-static discharge to which equipment may be sensitive. Observe precautions for handling electrostatic sensitive devices.
	Protective Earth (PE) terminal: Provided for connection of the protective earth (green or green/yellow) supply system conductor.
	Functional earth terminal: Used for non-safety purposes such as noise immunity improvement. NOTE: This connection shall be bonded to Protective Earth at the source of supply in accordance with national local electrical code requirements.
	Earth Ground. Functional earth connection. NOTE: This connection shall be bonded to Protective Earth at the source of supply in accordance with national and local electrical code requirements.
	Chassis Ground: Identifies a connection to the chassis or frame of the equipment shall be bonded to Protective Earth at the source of supply in accordance with national and local electrical code requirements.

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1. System Planning

1.1 Introduction

Platform overview

The WS360, WS370, and WS470 platforms provide PC-based functionality for the Experion PKS system and the TPS system, including Global User Stations (GUSs) and APP Nodes. For TPS systems, the platforms have an LCNP4 card installed allowing connection to the TotalPlant Process Network (TPN) coax through the LCN Media Access Unit (MAU). Additionally, these platforms connect to the Ethernet through a standard Ethernet adapter card or the optional FTE Ethernet adapter card.

Furniture options

The workstation may be mounted in a variety of Honeywell furniture. The following tables lists the types of furniture in which the computer can be mounted and indicates whether it may use a fixed rail mount or slide tray mount.

Furniture Type	Mounting Option	
	Fixed (rails)	Slide (tray)
MarkHon cabinet	Yes	Yes
Classic Console	Yes	Yes
Z Console	Yes	No
EZ Console	Yes	No
Icon Series Console	N/A	N/A

Software requirements

These platforms have been qualified with the Microsoft Windows 2000 Professional and XP Professional operating systems with Honeywell software. Refer also to the latest Software Change Notice (SCN) packaged with the platform for the specific Honeywell software that has been qualified for use with the WS360, WS370, and WS470.

BIOS configuration

The WS360 and WS370 platforms must have BIOS version A02 or later installed. The WS470 platform must have BIOS version A03 or later installed.

1.2 WS360 Description

WS360 enclosure

The WS360 can be positioned vertically or horizontally and used as a desktop, or mounted in the following Honeywell furniture:

- Classic Console
- Ergonomic (Z) Console
- Icon Series Console
- MarkHon Cabinet

When mounted in a cabinet, the enclosure uses the following amount of space:

- Slide Mount: 5U (8.75") of rack space
- Fixed Mount: 4U (7.00") of rack space



Figure 1-1 WS360 Enclosure for Cabinets and Consoles

Electronics module

The peripheral electronics assemblies in the WS360 are based upon either the Peripheral Component Interconnect (PCI) bus, AGP or USB 1.1. The standard DDR SDRAM memory for this platform is 1.0 GB ECC, optionally expandable to 2.0 GB ECC.

Storage and media devices

The standard mass storage for the WS360 is an 80 GB or larger hard drive. There are two separate EIDE interconnection cables: one for mass storage (hard disk) drive and the second for the CDRW/DVD+RW drive. Removable media devices in the Icon Console are connected via USB 1.1, and other removable media devices use a SCSI connection.

WS360 options

The WS360 comes with the following options:

Standard options

- Pentium 4 motherboard (single CPU)
- 512 KB Cache
- 1 AGP 8X Pro50 Slot
- Ports: 2 serial, 1 parallel, 8 USB 2.0 – 2 front & 6 back
- 4 PCI slots
- Dual bus Master IDE Ultra DMA 33/66/100
- Integrated Sound (Sound Blaster® Audigy II with onboard 1394)
- LAN (Intel PRO/1000 MTGigabit)
- 330 W Power Supply
- 2-512 MB DDR400, ECC SDRAM
- EIDE CDRW/DVDRW(+) Drive
- EIDE 80 GB or larger, 7200 RPM, Hard Drive
- 3.5" 1.44 Mb Floppy drive
- Dual Head AGP Video Graphics Card

Additional options

- LCNP4
- For Quad video – add a second dual video card (PCI)
- SCSI Controller
- Dual NIC for FTE
- Remote power/circuit breaker assembly
- Two additional 512 MB memory modules

WS360 specific slot requirements

The following table defines the slot requirements for WS360 configurations. Those devices identified as *optional* are only present if you specifically ordered them.

Table 1-1 WS360 Slot Configuration

Slot Identification	Device
<i>Single/Dual video configuration</i>	
AGP	Dual video card
PCI1	LCNP4
PCI2	Free slot
PCI3	SCSI (optional)
PCI4	Dual NIC used for FTE (optional)*
Empty slot	Remote power/circuit breaker assembly (optional)
<i>Quad video configuration</i>	
AGP	Dual video card
PCI1	LCNP4
PCI2	Dual video card
PCI3	SCSI controller (optional)
PCI4	Dual NIC used for FTE (optional)*
Empty slot	Remote power/circuit breaker assembly (optional)
* Dual NIC for FTE systems; requires additional configuration per FTE documentation.	

Memory configurations for WS360

The standard installed memory for the WS370 is 1 GB as shown in Figure 1-2. It can be expanded to 2 GB by installing two additional 512 MB memory modules in DIMM socket 1, as shown in Figure 1-3. The memory devices must be from the same memory supplier.

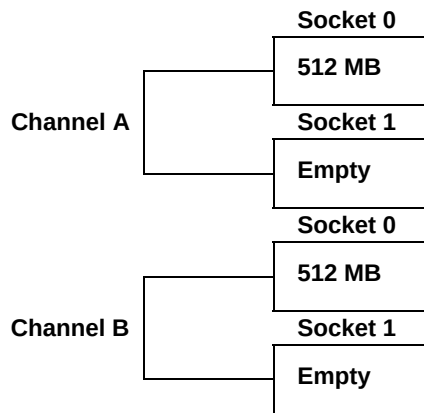


Figure 1-2 WS360 1 GB Memory Configuration

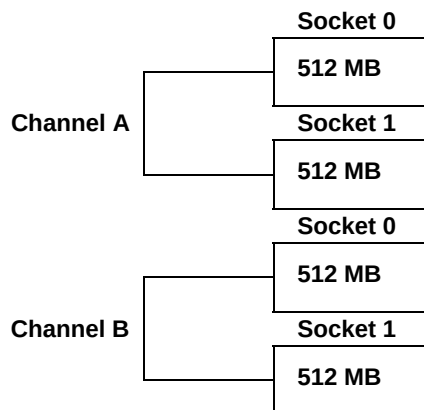


Figure 1-3 WS360 2 GB Memory Configuration

1.3 WS370 Description

WS370 enclosure

The WS370 can be positioned vertically or horizontally and used as a desktop, or mounted in the following Honeywell furniture:

- Classic Console
- Ergonomic (Z) Console
- Icon Series Console
- MarkHon Cabinet

When mounted in a cabinet, the enclosure uses the following amount of space:

- Slide Mount: 5U (8.75") of rack space
- Fixed Mount: 4U (7.00") of rack space



Figure 1-4 WS370 Enclosure for Cabinets and Consoles

Electronics module

The peripheral electronics assemblies in the WS370 are based upon either the Peripheral Component Interconnect (PCI) bus, AGP or USB 1.1/2.0 protocols; PCI Express and PCIX1x.

Storage and media devices

The standard mass storage for the WS370 is an 80 GB or larger hard drive. Except for the floppy drive, all mass storage and removable media devices connect using EIDE or SCSI depending on the configuration. The system has one EIDE connection for the CDRW/DVDRW(+) drive and four serial ATA (SATA) ports [0-3] for hard disk drives. For the Icon Console, external media devices connect to the USB port.

WS370 options

The WS370 comes with the following options:

Standard options

- Pentium 4 motherboard (single CPU)
- 800 megahertz front side bus
- 1 megabyte L2 Cache
- Ports: 1 to 2 serial ports (2nd is optional), 1 parallel, 8 USB 2.0 – 2 front & 6 back,
- Slots: 1 PCI Express x 1, 1 PCI Express x 16 Graphics, 4 PCI
- Integrated Sound (Sound Blaster® Audigy 2 with onboard 1394 port)
- LAN: Integrated PCI Express Broadcom® BCM5752 10/100/1000 gigabit Ethernet.
- 350 watts power factor correcting power supply
- 1 gigabyte 533 megahertz DDR2 ECC SDRAM
- EIDE DVD+RW drive with Sonic Record Now
- Serial ATA 80 gigabyte or larger, 7200 RPM, hard disk drive
- 3.5" 1.44 megabyte floppy drive
- ATI OpenGL V3100 dual video controller

Additional options

- LCNP4
- Serial port
- Matrox G450 quad PCI video display card
- SCSI Controller
- Dual NIC for FTE
- Remote power/circuit breaker assembly
- Additional 1 gigabyte memory modules

WS370 specific slot configurations

The following table defines the slot requirements for WS370 configurations. Those devices identified as *optional* are only present if you specifically ordered them.

Table 1-2 WS370 Slot Configuration

Slot Identification	Device
<i>Single/Dual video configuration</i>	
PCI express	Dual video card
PCI1	LCNP4
PCI2	Dual NIC (optional – used for FTE)*
PCI3	SCSI (optional)
PCIX1x	Remote power/circuit breaker assembly (optional)
PCI4	Serial port adapter
<i>Quad video configuration</i>	
PCI express	Serial port adapter
PCI1	LCNP4
PCI2	Dual NIC (optional – used for FTE)*
PCI3	SCSI (optional)
PCIX1x	Remote power/circuit breaker assembly (optional)
PCI4	Quad video card
* Dual NIC for FTE systems; requires additional configuration per FTE documentation.	

Memory configurations for WS370

The standard installed memory for the WS370 is 1 GB as shown in Figure 1-5. It can be expanded to 3 GB by installing additional memory modules in the appropriate sockets as shown in Figure 1-6. The memory devices must be from the same memory supplier.

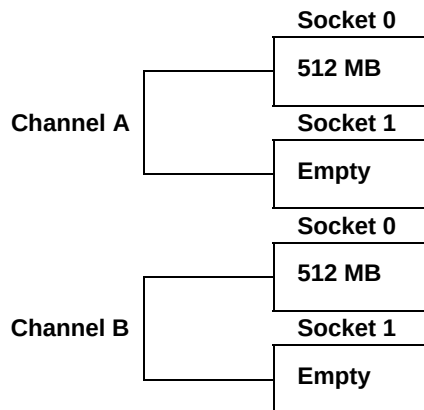


Figure 1-5 WS370 1 GB Memory Configuration

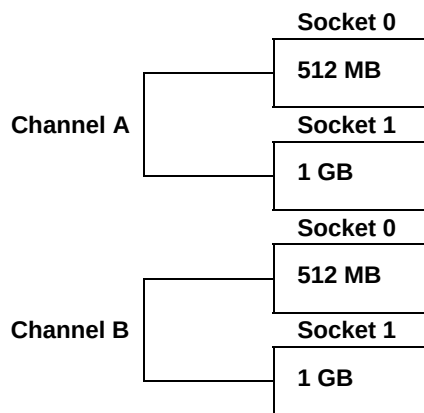


Figure 1-6 WS370 3 GB Memory Configuration

1.4 WS470 Description

WS470 enclosure

The WS470 can be positioned vertically or horizontally and used as a desktop, or mounted in the following Honeywell furniture:

- Classic Console
- Ergonomic (Z) Console
- Icon Series Console
- MarkHon Cabinet

When mounted in a cabinet, the enclosure uses the following amount of space:

- Slide Mount: 5U (8.75") of rack space
- Fixed Mount: 4U (7.00") of rack space



Figure 1-7 WS470 Enclosure for Cabinets and Consoles

Electronics module

The peripheral electronics assemblies in the WS470 are based upon either the Peripheral Component Interconnect (PCI) bus, PCI Express or PCIX or USB 1.1/2.0 protocols.

Storage and media devices

The standard mass storage for the WS470 is an 80 GB or larger SATA hard drive. Except for the floppy drive, all mass storage and removable media devices connect using EIDE or SCSI depending on the configuration. The system has two EIDE connections for the CDRW/DVDRW(±) drive and two serial ATA (SATA) ports for mass storage. For the Icon Console, external media devices connect to the USB 2.0 ports.

WS470 options

The WS470 comes with the following options:

Standard options

- Xeon motherboard (single or dual CPU)
- 800 megahertz front side bus
- 2 megabyte or greater L2 Cache
- Ports: 2 serial ports (9-pin) , 1 parallel (DB25), 8 USB 2.0 – 2 front & 6 back,
- Slots: 1 PCI Express x 4, 1 PCI Express x 16, 1 32-bit/33 MHz PCI, and 1 64-bit/100 MHz PCI-X
- Integrated Sound (Sound Blaster® emulation))
- LAN: Integrated PCI Express 64-bit 10/100/1000 Mps Ethernet
- 550 watts power factor correcting power supply (auto ranging)
- 1 gigabyte (2x512MB) 400 megahertz DDR2 ECC SDRAM
- EIDE CDRW/DVDRW (±) drive with Sonic Record Now
- Serial ATA 80 gigabyte or larger, 7200 RPM, hard disk drive
- 3.5" 1.44 megabyte floppy drive
- ATI FireGL V3100 dual video controller (128 MB)

Additional options

- LCNP4M (mid-size)
- Serial port
- Matrox QID quad video display card
- SCSI Controller
- Dual NIC for FTE
- Single NIC for Workstation
- Remote power/circuit breaker assembly
- Additional 1 gigabyte memory modules

WS470 specific slot configurations

The Single/Dual video display option is the default option for the WS470 platform. The Quad video display option includes three screen and quad screen video capabilities. The following tables define the slot requirements for WS470 configurations for TPS, FTE, and Experion systems. Those devices identified as *optional* are only present if you specifically ordered them.

Table 1-3 WS470 Slot Configuration for TPS Systems

Slot Identification	Device
<i>Single/Dual video configuration</i>	
PCIX – Slot 4	SCSI - Optional
PCIe x 4 – Slot 3	Empty or power adapter harness
PCI – Slot 2	LCNP4M
PCIe x 16 – Slot 1	ATI – FireGL Dual Video Card
<i>Quad video configuration</i>	
PCIX – Slot 4	SCSI - Optional
PCIe x 4 – Slot 3	Empty or power adapter harness
PCI – Slot 2	LCNP4M
PCIe x 16 – Slot 1	MATROX QID Quad video card

Table 1-4 WS470 Slot Configuration for Experion/FTE Systems

Slot Identification	Device
<i>Single/Dual video configuration</i>	
PCIX – Slot 4	Single / Dual Intel Pro 1000 Ethernet
PCIe x 4 – Slot 3	Empty or power adapter harness
PCI – Slot 2	LCNP4M
PCIe x 16 – Slot 1	ATI – FireGL Dual Video Card
<i>Quad video configuration</i>	
PCIX – Slot 4	Single / Dual Intel Pro 1000 Ethernet
PCIe x 4 – Slot 3	Empty or power adapter harness
PCI – Slot 2	LCNP4M
PCIe x 16 – Slot 1	MATROX QID Quad video card

Memory configurations for WS470

The standard installed memory for the WS470 is 1 GB as shown in Figure 1-8. It can be expanded to 3 GB by installing additional memory modules in the appropriate sockets as shown in Figure 1-9. The memory devices must be from the same memory supplier, ECC, single rank, registered, X8 organization.

DIMM Socket	Memory Size
1*	512 MB
2	512 MB
3	Empty
4	Empty
5	Not Used
6	Not Used
* DIMM Socket 1 is closest to the edge of the mother board	

Figure 1-8 WS470 1 GB Memory Configuration

DIMM Socket	Memory Size
1	512 MB
2	512 MB
3	1 GB
4	1 GB
5	Not Used
6	Not Used
* DIMM Socket 1 is closest to the edge of the mother board	

Figure 1-9 WS470 3 GB Memory Configuration

1.5 Finding Information for Your Computer

Dell documentation

The following table lists Dell publications and other sources of information that will be useful when installing, operating or servicing the computer.

Table 1-5 Dell Publications

Publication	Contains information on	Available
Readme files	Last-minute updates about technical changes to your computer or advanced technical-reference material for experienced users or technicians.	Drivers and utilities CD
System Information Label	System board connectors Location of system board components	Located on the inside cover of your computer.
<i>Dell System Information Guide</i>	Warranty information Safety information	Packaged with the computer www.dell.com
<i>Setup and Quick Reference Guide</i>	How to set up my computer How to care for my computer Troubleshooting information How to run the Dell Diagnostics How to open my computer cover How to locate other documentation	Packaged with the computer www.dell.com
<i>Dell Precision™ Workstation 360 User's Guide</i> <i>Dell Precision™ Workstation 370 User's Guide</i> <i>Dell Precision™ Workstation 470 User's Guide</i>	How to remove and replace parts Technical specifications How to configure system settings How to troubleshoot and solve problems	Product Documentation CD www.dell.com
The <i>Dell Precision™ 360 Service Manual</i> or the Service Manual for the 370 or 470.	Details on computer configuration How to remove and replace parts	Product Documentation CD www.dell.com

1. System Planning

1.5. Finding Information for Your Computer

Honeywell documentation

Honeywell's computer platforms can be configured in a number of ways and installed in various consoles and cabinets. The following table lists other Honeywell publications that may be useful when installing or operating your system.

Table 1-6 Honeywell Publications

Publication	Contains information on
HC05: <i>Planning, Installation and Service for Honeywell Icon Series Console</i>	Includes instructions for mounting the computer in the Icon console.
US13600 or later: <i>Classic Console Service</i>	Includes information on the Classic console furniture and peripheral devices.
US13520a or later: <i>Ergonomic Classic Console Service</i>	Includes information on the Z and EZ console furniture and peripheral devices.
ADP01: <i>Honeywell Peripheral Adapters</i>	Contains information for using the OEP/IKB adapter with computer platforms that do not have the ISA card.
RE01: <i>Remote Peripheral Systems Installation and Upgrade</i>	Contains information for using Honeywell's short-haul remote (100 feet) option.
RE02: <i>Long-Haul Remote Systems Installation and Upgrade</i>	Contains information for using Honeywell's long-haul remote (300 feet) option.
FE05: <i>Fault Tolerant Ethernet Installation and Service Guide</i>	Contains information for installing and using FTE on a TPS or Experion PKS node.

1.6 Specifications

Environmental specifications for cabinets and consoles

The following table lists allowable operating environmental limitations for cabinets and consoles.

Description	Cabinets	Consoles
Ambient room temperature	5 – 30 deg C	5 – 40 deg C
Humidity	20 – 80% RH, non-condensing	20 – 80% RH, non-condensing
Operating vibration	0.012" P-P displ to 12.7 Hz, then 0.1g to 150 Hz	0.012" P-P displ to 20 Hz, then 0.25g to 150 Hz
Site induced shock	6g, 10 msec half-sine	10g, 10 msec half-sine

WS360 system specifications

Table 1-7 Specifications for WS360

Description	Specification
<i>Operating power requirements</i>	
AC Voltage	120 (90-135) Vrms 240 (180-265) Vrms
AC RMS Current	1.0 Arms 0.6 Arms
AC Power	330 Watts Max
<i>Weight and dimensions</i>	
Height	165.1 mm
Width	444.5 mm
Depth	457.2 mm
Weight	14.0 Kg depending on options
<i>Hard disk drive specifications</i>	
DC 5 volt Power (typ)	5 Watts +/- 5%
DC 12 volt Power (typ)	12 Watts +10%, -8%
Other DC POWER	22 Watts Max, 10.0 Watts idle
AC Power	N/A
Height	25.4 mm
Width	101.6 mm
Depth	146.0 mm
Weight	0.6 kg

1. System Planning

1.6. Specifications

WS370 system specifications

Table 1-8 Specifications for WS370	
Description	Specification
<i>Operating power requirements</i>	
AC Voltage	120 (90-135) Vrms 240 (180-265) Vrms
AC RMS Current	1.0 Arms 0.6 Arms
AC Power	330 Watts Max
<i>Weight and dimensions</i>	
Height	165.1 mm
Width	444.5 mm
Depth	457.2 mm
Weight	14.0 Kg depending on options
<i>Hard disk drive specifications</i>	
DC 5 volt Power (typ)	5 Watts +/- 5%
DC 12 volt Power (typ)	12 Watts +10%, -8%
Other DC POWER	22 Watts Max, 10.0 Watts idle
AC Power	N/A
Height	25.4 mm
Width	101.6 mm
Depth	146.0 mm
Weight	0.6 kg

WS470 system specifications

Table 1-9 Specifications for WS470	
Description	Specification
<i>Operating power requirements</i>	
AC Voltage	120 (90-135) Vrms 240 (180-265) Vrms
AC RMS Current	1.0 Arms 0.6 Arms
AC Power	550 Watts Max
<i>Weight and dimensions</i>	
Height	167.4 mm
Width	447.04 mm
Depth	444.5 mm
Weight	18.6 Kg depending on options
<i>Hard disk drive specifications</i>	
DC 5 volt Power (typ)	5 Watts +/- 5%
DC 12 volt Power (typ)	12 Watts +10%, -8%
Other DC POWER	22 Watts Max, 10.0 Watts idle
AC Power	N/A
Height	25.4 mm
Width	101.6 mm
Depth	146.0 mm
Weight	0.6 kg

1. System Planning

1.6. Specifications

Removable media specifications

The following table lists specifications for removable media that can be used with the WS360, WS370 or WS470 platforms.

Table 1-10 Specifications for Removable Media

Removable Media Power Requirements				
Description	Device Requirements			
	CDRW/DVDRW	Zip drive	DAT drive	Mini cartridge tape drive
DC 5 volt Power	6.0 Watts	5.0 Watts	4.0 Watts	5 Watts
DC 12 volt Power	21.6 Watts	N/A	5 Watts	3 Watts
Other DC POWER	N/A	N/A	N/A	N/A
AC Power	N/A	N/A	N/A	N/A
AC RMS Current	N/A	N/A	N/A	N/A
AC Power	N/A	N/A	N/A	N/A
Height	40.64 mm	41.5 mm	40.6 mm	42.5 mm
Width	146.05 mm	148 mm	148 mm	148 mm
Depth	14.22 mm	163 mm	1456.5 mm	152 mm
Weight	1.20 kg	0.67 kg	1.09 kg	0.46 kg

Additional specifications

Refer to the specific computer user guide for additional technical specifications and the vendor documentation for specifications on the peripheral devices.

Regulatory and safety compliance



WARNING

Honeywell does not claim Safety Compliance or Electromagnetic Compatibility (EMC) Compliance for system equipment configurations that have not been described in this manual as standard system configurations. Any equipment configuration other than that described in this publication decertifies the Safety and EMC compliance of this product.

2. Platform Installation

2.1 Introduction

Overview

This section contains procedures for installing and cabling the WS360, WS370, and WS470 in Honeywell consoles and cabinets. Refer also to your specific console documentation as listed in Table 1-6.

Installation tasks

The specific tasks you need to perform will vary depending upon the type of furniture in which you are installing the computer. The following table lists the major platform installation tasks.

Table 2-1 System Installation Tasks		
Task		For more information see . . .
✓	Be aware of all power and grounding requirements for your furniture.	Specific site requirements 2.2 "Power and Grounding Requirements" <i>TPN System Installation (SW20-600)</i> , Section 4, "System Grounding"
✓	Verify pinning for external media drives is correct	2.3 "Pinning Media Drives"
✓	Install the computer in the furniture	2.4 "Installing the Computer in Consoles" 2.5 "Installing the Computer in Cabinets"
✓	Connect device cables	2.6 "Connecting Cables to a WS360" 2.7 "Connecting Cables to a WS370" 2.8 "Connecting Cables to a WS470"
✓	If you are remoting the computer, connect the remote components.	For short haul remote: <i>Remote Peripheral Systems Installation and Upgrade (RE01)</i> For long haul remote: <i>Long-Haul Remote Systems Installation and Upgrade (RE02)</i>

2.2 Power and Grounding Requirements

Grounding for PC based nodes

The ground connection is made through the third wire in the AC power cord.

Grounding consoles and cabinets

The *TPN System Installation* manual, Section 4, “System Grounding” contains information on grounding furniture, including the following:

- Ground Wiring Overview
- Grounding LCN Cabinets and Stations
- Cabinet Logic Ground
- Grounding LCN Cables
- Grounding Stations in Z-Console Furniture

AC Power Warning



WARNING

The power supply circuit is connected to AC power when the power cable is connected. The power control switch on the front panel only enables the power supply circuit outputs.



ATTENTION

It is strongly recommended that the power cord be connected to a clean power source with backup such as an Uninterruptible Power Source (UPS).

Selecting the correct power setting

The WS370 platform has a voltage selection switch next to the AC power cord connector as it does not automatically sense the power supply. Before connecting power, verify the switch is in the correct position for the type of power you are using. The WS360 and WS470 platforms automatically sense the power supply therefore there is no positioning switch to be set.



2.3 Pinning Media Drives

Configure SCSI address for removable media drives

The following table lists the SCSI address assignments for any removable media drives in your system for the WS360 and WS370. The SCSI card is optional for the WS470 if you are not using FTE. The WS470 interfaces to SCSI media through the USB to SCSI adapter.

Table 2-2 Address Assignments for Media Drives

Drive Type	Address Assignment
Tape drive	SCSI address 6
Second tape drive	SCSI address 4
ZIP disk drive	SCSI address 4
Second ZIP drive	SCSI address 6
CD/DVD-ROM	SCSI address 5
Note: If these unit numbers conflict, use address 3, then 2.	

2. Platform Installation

2.4. Installing the Computer in Consoles

2.4 Installing the Computer in Consoles

Overview

This section contains the following procedures for installing the WS360, WS370, and WS470 in console furniture:

- Install WS360 or WS370 computer in Icon console
- Install WS470 computer in Icon console
- Install computer in Z console
- Install computer in Classic console using the fixed mount option
- Install computer in Classic console using the slide mount option

Install WS360 or WS370 computer in Icon console

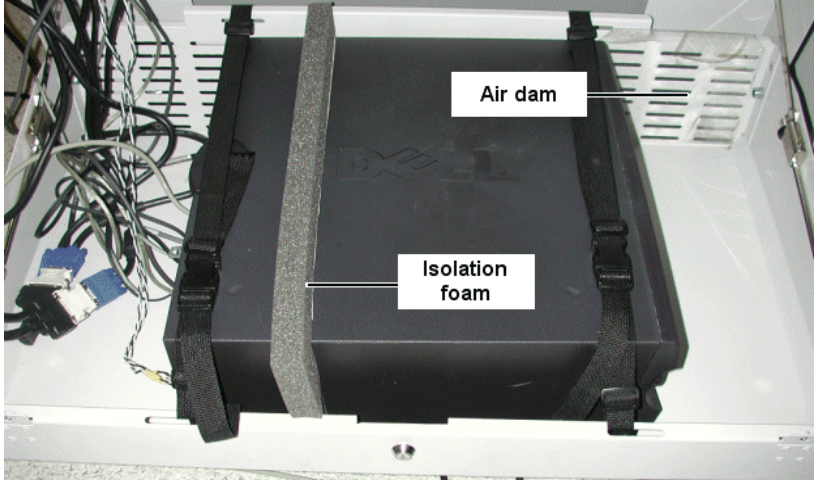
Use this procedure to install a WS 360 or a WS370 system platform and add the air dam and isolator foam.

If you are installing a WS470 computer in an Icon console refer to the “Install WS470 computer in Icon console” procedure on page 40.

Step	Action
1	Place the computer in the electronic enclosure with the rear of the computer facing the left side of the enclosure, making sure the computer is centered.
2	Fasten the tie-down straps and tighten.
3	Remove the adhesive from the isolation foam and place it on top of the computer slightly left of center.

2. Platform Installation

2.4. Installing the Computer in Consoles

Step	Action
4	Remove the adhesive from the air dam, and use it to cover the bottom ventilation slots on the right side of the electronic enclosure. 
5	If necessary, adjust the tension for raising and lowering the electronic enclosure by loosening or tightening the flat head screws on each side of the mounting assembly.
6	Go to the following relevant section: • Section 2.6, "Connecting Cables to a WS360". • Section 2.7, "Connecting Cables to a WS370".

2. Platform Installation

2.4. Installing the Computer in Consoles

Install WS470 computer in Icon console

Before you begin

The WS470 platform is positioned in the Icon electronics enclosure with the cable connections on the right side (as seen from the console rear). Before installing the WS470, reroute all your cables to the right side (as seen from the console rear) of the console as shown in this figure.



WS470 in electronics enclosure

The following figure shows the WS470 platform mounted in the Icon console's electronic enclosure.

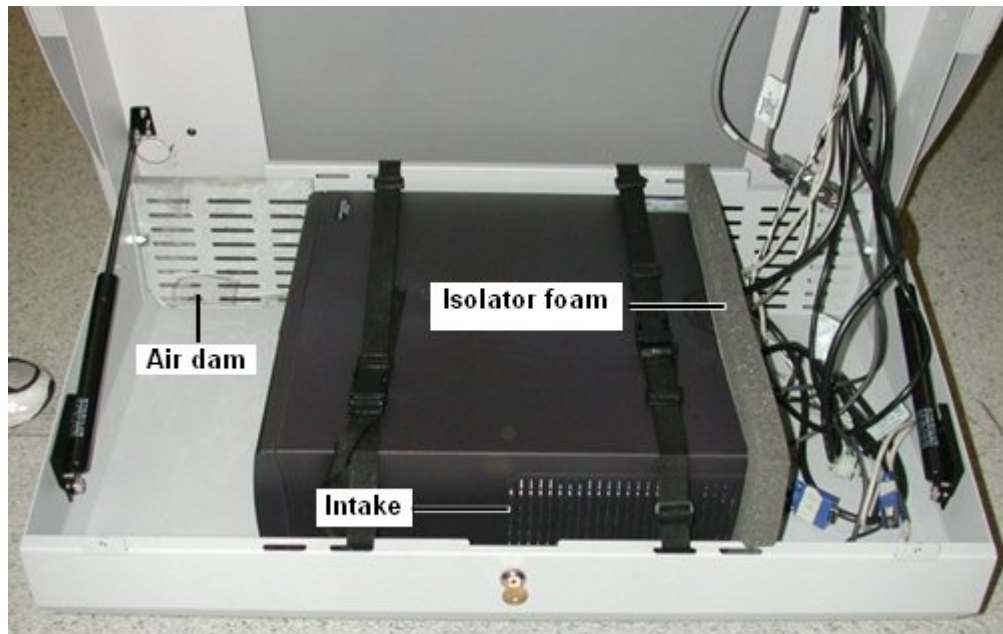


Figure 2-1 WS470 in Electronics Enclosure

Install WS470 in electronics enclosure

Use the following procedure to install the system platform and add the air dam and isolator foam.

Step	Action
1	Place the computer in the electronic enclosure with the back of the computer facing the right side (as seen from the console rear) of the enclosure making sure the computer is centered. See Figure 2-1.
2	Fasten the tie-down straps and tighten.
3	Remove the adhesive from the isolation foam and place the isolation foam on top of the computer to the right of the intake vents. See Figure 2-1.

2. Platform Installation

2.4. Installing the Computer in Consoles

Step	Action
4	Remove the adhesive from the air dam, and use it to cover the ventilation slots on the left side (as seen from the console rear) of the electronic enclosure. See Figure 2-1.
5	If necessary, adjust the tension for raising and lowering the enclosure by loosening or tightening the flat head screws on each side of the mounting assembly.

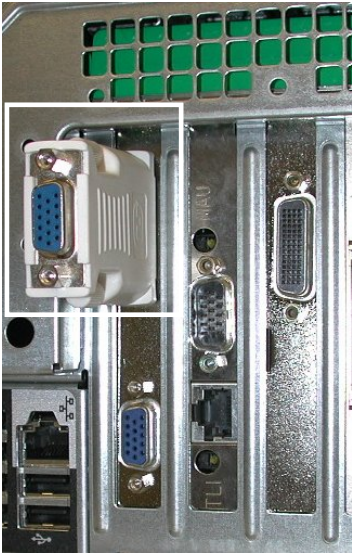
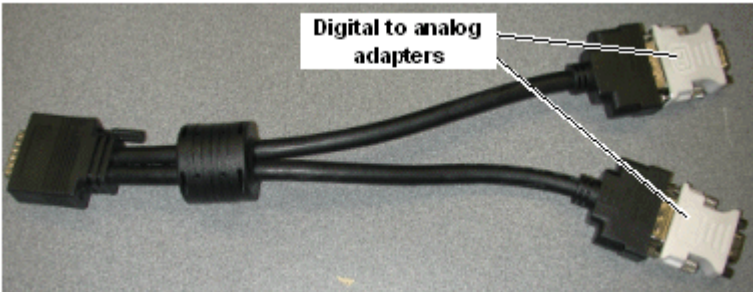
Connect system cables

Use the following procedure to connect the cables when you are installing the platform in an Icon console.

Step	Action
1	Connect the mouse and keyboard cables to the PS/2 connectors.
2	Connect the speaker cable to the audio port.
3	If you are not using Fault Tolerant Ethernet (FTE), connect the Ethernet cable to the RJ-45 connector on the Network Interface Card. If you are using FTE, you will need to connect the FTE cables according to the instructions in the <i>FTE Installation and Service Guide</i>.

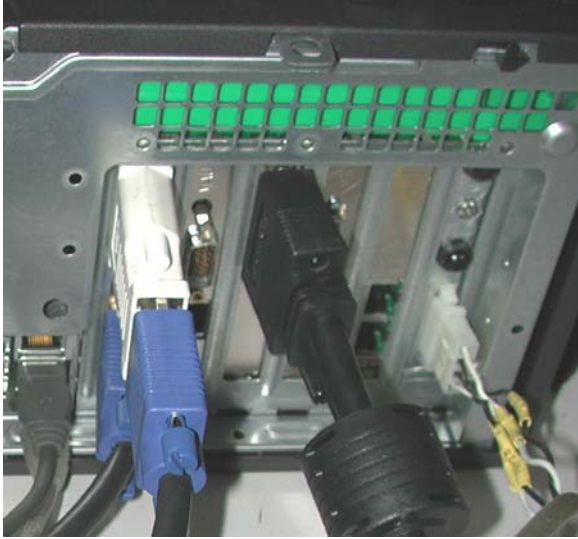
2. Platform Installation

2.4. Installing the Computer in Consoles

Step	Action
4	Attach the digital to analog converter to the digital connector on the video card as shown. 
5	Connect the DVI-D to DVI-A video cables to the analog video connectors on the video card. If your console does not have more than two monitors, go to step 8.
6	Connect the digital to analog adapters to the Y cable as shown. 

2. Platform Installation

2.4. Installing the Computer in Consoles

Step	Action
7	Connect the Y cable to the video connector on the second video card (in the PCI slot). 
8	Connect the monitor video cables to the computer according to the information in "Connect remaining WS470 cables and power" in section 2.8, Connecting Cables to a WS470 on page 84.
9	If your console has an IKB or an OEP, connect the cables according to the instructions in the ADP01: <i>Honeywell Peripheral Adapters</i>
10	Connect the LCN MAU cable to the MAU connector on the LCNP4M card if one is present.
11	Connect the USB Hub's upstream cable to the USB port on the system chassis.
12	Connect the USB Hub's power cable to the system chassis.
13	Connect the disk power cable to the four-pin socket.
14	Secure any loose cables, and verify that all cables have proper strain relief.

Add computer power cable

If your console has the local electronics option, use this procedure to add the computer's power cable.



ATTENTION

Before you begin this procedure, you will need to TURN OFF POWER to the console and make sure the console base and console legs are accessible.

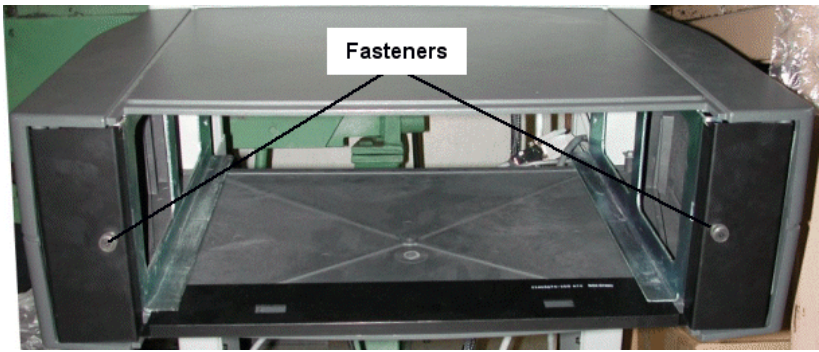
Step	Action
1	Connect the power cable to the computer, routing the cable through the electronics box lower cable access hole and into the lower leg area and then to the power entry box.
2	Plug the computer power cable into the console power entry box.
3	Replace the console foot treadle, and tighten the flange nuts.
4	Plug in the console AC power cable, or restore power at the main switch.
5	Turn the console AC power switch to ON.

2. Platform Installation

2.4. Installing the Computer in Consoles

Install computer in Z console

The computer is installed in a fixed 15-degree reclining position in the upper or lower furniture bay of Z and EZ Console furniture. Use this procedure to install the WS360, WS370, or WS470 computer in the Z console furniture bay.

Step	Action
1	From the rear of the console, loosen the two captivated fasteners on the two mounting bracket covers and remove the bracket covers. 
2	Slide the furniture bay cover off the mounting railings.

2. Platform Installation
2.4. Installing the Computer in Consoles

Step	Action
3	Verify that the console mounting rails are clear of any installed cables.  A photograph showing the interior of a console. Two green metal mounting rails are visible, extending from the top of the console down to a black base. Several cables are bundled together and run along the right side of the rails. The background shows a white wall and a cardboard box with the Honeywell logo.

2. Platform Installation

2.4. Installing the Computer in Consoles

Step	Action
4	Place the back of the computer on the mounting railings, and slide the computer into the furniture bay until the front of the computer clears the retaining lip of the horizontal bracket.  

2. Platform Installation
2.4. Installing the Computer in Consoles

Step	Action
5	Position the furniture bay cover back on the mounting railings and slide it all the way back.  A photograph showing a grey, wedge-shaped furniture bay cover being slid back onto a metal mounting railing. The cover is positioned at an angle, and its front edge is being guided into the railing. The background shows a white console structure and some internal wiring.
6	Replace the two mounting bracket covers and tighten the captivated fasteners.  A photograph of a Dell computer system installed in a console. The system includes a monitor, a tower unit, and a keyboard. Two black mounting bracket covers are visible on the sides of the console, and the Dell logo is prominently displayed on the front of the tower unit. The console is mounted on a stand, and various cables are visible at the bottom.

2. Platform Installation

2.4. Installing the Computer in Consoles

Step	Action
7	From the front of the console, remove the front cover that is directly under the keyboard by turning the fasteners counter-clockwise and lift the cover off the mounting brackets. 
8	Go to the following relevant section: • Section 2.6, "Connecting Cables to a WS360". • Section 2.7, "Connecting Cables to a WS370". • Section 2.8, "Connecting Cables to a WS470".

Classic Console mounting requirements

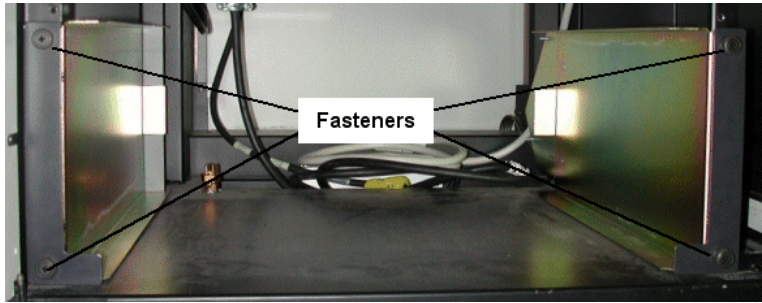
The WS360, WS370 and WS470 is installed in a horizontal position in the Classic Console using a slide mount or fixed mount option. The following table describes the requirements for these mounting options.

Table 2-3 Slide and Fixed Mounting Requirements

Option	Rack Space Needed	Maximum WS360s	Mounting Requirements
Slide	5U (8.75 inches)	1 per console	Must be mounted on bottom of console The 3U of unused space must have a 3U blank front panel and air duct baffle installed.
Fixed	4U (7.00 inches)	2 per console	If only one WS360 is installed, it must be mounted on the bottom of console with a 4U blank front panel and air duct baffle installed in the unused space.

Install computer in Classic console using the fixed mount option

Use this procedure to install the WS360, WS370 or WS470 computer in Classic console furniture using the fixed mount option.

Step	Action
1	Remove the console door to access the console enclosure.
2	Determine an appropriate location for the computer by referring to the requirements listed in Table 2-3.
3	If the retaining brackets are attached, loosen the two captivated fasteners on each retaining bracket and remove them. 
4	Verify that the console mounting rails are clear of any installed cables. 

2. Platform Installation

2.4. Installing the Computer in Consoles

Step	Action
5	Slide the computer all the way into the console enclosure between the mounting railings as shown. 
6	Replace the two retaining brackets and tighten the four captivated fasteners. 

2. Platform Installation

2.4. Installing the Computer in Consoles

Step	Action
7	If you are installing a second computer in the console, repeat steps 2 through 6 for the second computer. Attach the blank panel to the console rails using two hex head screws and two washers, making sure the flat side of the washer touches the screw head and the washer teeth touch the panel.  Note: Installing a blank panel ensures that the airflow within the console allows proper cooling of the computer.
8	Go to the following relevant section: • Section 2.6, "Connecting Cables to a WS360". • Section 2.7, "Connecting Cables to a WS370". • Section 2.8, "Connecting Cables to a WS470".

Install computer in Classic console using the slide mount option

Use this procedure to install the WS360, WS370 or WS470 computer in Classic console furniture using the slide mount option.

Step	Action
1	Remove the console door to access the console enclosure.
2	If the retaining brackets are attached, loosen the two captivated fasteners on each retaining bracket and remove them.
3	Verify that the slide mount bottom and sides are clear of any installed cables.
4	Using the mounting tray handle, fully extend the tray from the console.
5	Place the computer on the mounting tray and slide the tray back into the console.

2. Platform Installation

2.4. Installing the Computer in Consoles

Step	Action
6	Replace the two retaining brackets and tighten the four captivated fasteners.
7	Attach the blank panel to the console rails using two hex head screws and two washers, making sure the flat side of the washer touches the screw head and the washer teeth touch the panel.  Note: Installing a blank panel ensures that the airflow within the console allows proper cooling of the computer.
8	Go to the following relevant section: • Section 2.6, "Connecting Cables to a WS360". • Section 2.7, "Connecting Cables to a WS370". • Section 2.8, "Connecting Cables to a WS370".

2.5 Installing the Computer in Cabinets

Overview

The WS360, WS370, and WS470 computer mounts horizontally in a cabinet. The human interface is accessed locally in the cabinet or remotely using Honeywell's remote option.

Mounting devices in cabinet

The various devices mounted in the cabinet require RETMA standard cabinet openings as specified in the following table.

Option	Rack Space Needed	Mounting Requirements
Slide mounted WS360	5U 8.75 inches	Any unused space must have a blank front panel and air duct baffle installed in the unused space to ensure proper airflow in the cabinet. Due to thermal constraints, only 6 total computers per cabinet are allowed
Fixed mounted WS360	4U 7.00 inches	
Human interface: 4 port KVM switch; 17-inch CRT; keyboard & cursor device	12 U 21 inches	Mount in rack space 13U through 24U to ensure the keyboard is at the proper height
Human interface: 8 port KVM switch; 15-inch LCD; keyboard & cursor device	1U 1.88 inches	Mount in rack space 13U on telescopic slides

Unused cabinet spaces



ATTENTION

To ensure proper computer cooling and airflow through the cabinet, all unused rack mount locations must have a blank front panel and air duct baffle installed.

2. Platform Installation

2.5. Installing the Computer in Cabinets

Example rack mounting locations using slide mounts

The following table is an example of a cabinet configuration using the slide mount option. The **Rack Mount** column lists the RETMA standard units in the cabinet.

Component Description	Rack Mount
Long-haul remote option	40U
5U high slide mount for workstation (with long haul remote option)	39U
	38U
	37U
	36U
	35U
5U high slide mount kit for workstation (with short-haul remote option)	34U
	33U
	32U
	31U
	30U
5U high slide mount for workstation (with short-haul remote option)	29U
	28U
	27U
	26U
	25U
12U enclosure for 17-inch monitor, keyboard, mouse and 4-port KVM switch	24U
	23U
	22U
	21U
	20U
	19U
	18U
	17U
	16U
	15U
	14U
	13U
6U high slide mount for Dell PE1800 server (KVM human interface option)	12U
	11U
	10U
	9U
	8U
6U high slide mount for Dell PE1800 server (KVM human interface option))	7U
	6U
	5U
	4U
	3U
	2U
	1U

Note: Short haul remote electronic box mounts on brackets that attach to the rear of the computer. Long haul remote box uses 1U of space mounted directly above computer.

Example rack mounting locations using fixed mounts

Using all fixed mounts, up to two Dell PE1600SC servers may be mounted in the first 10U of rack mounting space, starting at the bottom of the cabinet. The following table is an example of a cabinet configuration using the fixed mount option. The **Rack Mount** column lists the RETMA standard units in the cabinet.

Component Description	Rack mount
4U high fixed mount for workstation (with short haul remote option)	40U
	39U
	38U
	37U
4U high fixed mount for workstation (with short haul remote option)	36U
	35U
	34U
	33U
4U high fixed mount for workstation (with short haul remote option)	32U
	31U
	30U
	29U
4U high fixed mount for workstation (with short haul remote option)	28U
	27U
	26U
	25U
12U enclosure for 17-inch monitor, keyboard, mouse and 4-port KVM switch	24U
	23U
	22U
	21U
	20U
	19U
	18U
	17U
	16U
	15U
	14U
	13U
	12U
	11U
5U high fixed mount for Dell PE1800 server (KVM human interface)	10U
	9U
	8U
	7U
	6U
5U high fixed mount for Dell PE1800 server (KVM human interface)	5U
	4U
	3U
	2U
	1U

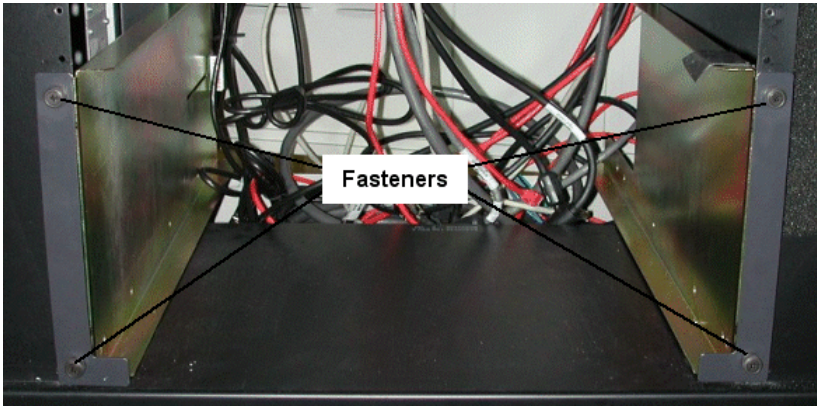
Note: Short haul remote electronic box is mounted on brackets that attach to the rear of the computer.

2. Platform Installation

2.5. Installing the Computer in Cabinets

Install computer in cabinet using the fixed mount option

Use this procedure to install the computer in the cabinet using the fixed mount option.

Step	Action
1	Open the cabinet door to access the inside of cabinet.
2	If the retaining brackets are attached, loosen the two captivated fasteners on each bracket and remove them. 
3	Verify that the cabinet mounting rails are clear of any installed cables.
4	Slide the computer all the way into the cabinet enclosure between the mounting railings.
5	Replace the two mounting bracket covers and tighten the four screws.
6	Go to the "Install air duct baffles and blank panel front covers in cabinet" procedure in this section.

Install computer in cabinet using the slide mount option

Use this procedure to install the computer in the cabinet using the slide mount option.

Step	Action
1	Open the cabinet door to access the inside of cabinet.
2	If the retaining brackets are attached, loosen the two captivated fasteners on each bracket and remove them.
3	Verify that the slide mount bottom and sides are clear of any installed cables.
4	Using the mounting tray handle, pull the tray to its fully extended and locked position. 

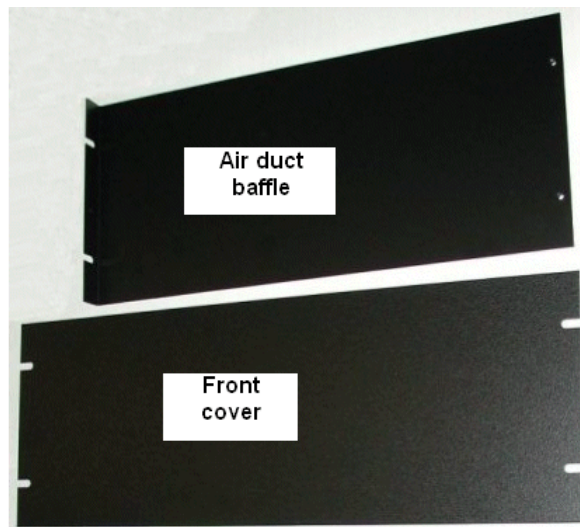
2. Platform Installation

2.5. Installing the Computer in Cabinets

Step	Action
5	Place the computer on the mounting tray so the bottom surface of the front bezel is aligned with the front surface of the slide tray and slide the tray back into the console. 
6	Replace the two retaining brackets and tighten the four captivated fasteners.
7	From the rear of the cabinet, slide the PC retainer bracket (L-shaped angle bracket) forward in its slot until it contacts the back surface of the PC chassis. Tighten the lock nut.
8	Go to the "Install air duct baffles and blank panel front covers in cabinet" procedure in this section.

Install air duct baffles and blank panel front covers in cabinet

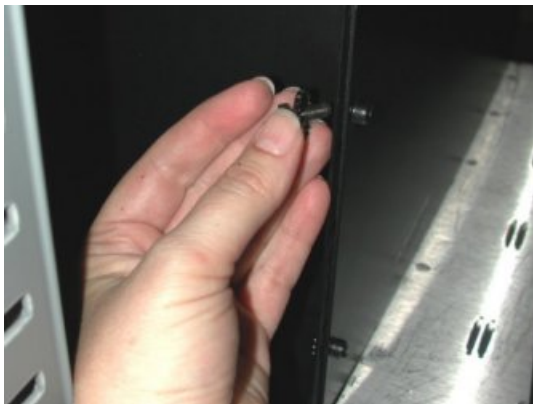
Use this procedure to install the air duct baffle and blank panel front cover in the cabinet. Installing these parts will insure that the airflow within the cabinet allows proper cooling of the computers.



Step	Action
1	Place the air duct baffle inside the cabinet with the bent tab resting along the front of the right cabinet rail as shown in the picture.  A photograph showing the air duct baffle installed inside a cabinet. The baffle is positioned horizontally, with its bent tab resting against the right cabinet rail. The front cover is visible below it.

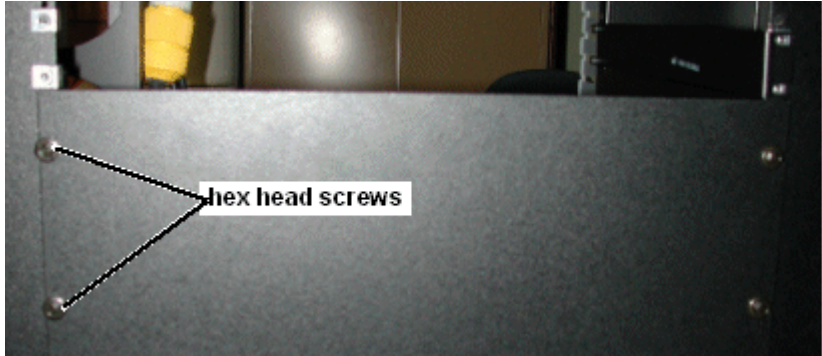
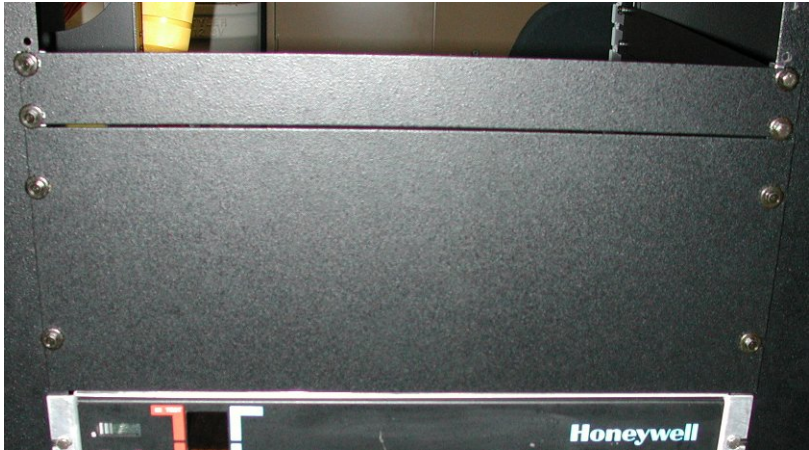
2. Platform Installation

2.5. Installing the Computer in Cabinets

Step	Action
2	Attach the air duct baffle to the rear cabinet rail using two hex head screws and two washers, making sure the flat side of the washer touches the screw head and the washer teeth touch the cabinet rail. 
3	Place the blank panel across the front of the cabinet rails and attach it to the air duct baffle and right cabinet rail using two hex head screws and two washers; make sure the flat side of the washer touches the screw head and the washer teeth touch the panel. 

2. Platform Installation

2.5. Installing the Computer in Cabinets

Step	Action
4	Attach the other side of the blank panel to the left cabinet rail using two hex head screws and two washers, making sure the flat side of the washer touches the screw head and the washer teeth touch the panel. 
5	Place the 1 blank panel across the front of the cabinet rails and attach it to the air duct baffle and right cabinet rail using two hex head screws and two washers; make sure the flat side of the washer touches the screw head and the washer teeth touch the panel. 

2. Platform Installation

2.5. Installing the Computer in Cabinets

Step	Action
6	Attach the other side of the blank panel to the left cabinet rail using two hex head screws and two washers, making sure the flat side of the washer touches the screw head and the washer teeth touch the panel. 
7	Go to the following relevant section: • Section 2.6, "Connecting Cables to a WS360". • Section 2.7, "Connecting Cables to a WS370". • Section 2.8, "Connecting Cables to a WS470".

Installing optional components

Honeywell provides a number of other options that may be installed in the cabinet with your computer. Each of the following options is packaged with installation instructions.

Component	Description
Short-haul remote components	Used for remoting the computer up to 100 feet from the cabinet. See <i>Remote Peripheral Systems Installation and Upgrade (RE01)</i>
OEP/IKB adapter	Used to connect an OEP or IKB that uses an ISA connection to the computer's serial port. See <i>Honeywell Peripheral Adapters (ADP01)</i>
Long-haul remote components	Used for remoting the computer up to 300 feet from the cabinet. See <i>Long-Haul Remote Systems Installation and Upgrade (RE02)</i> .

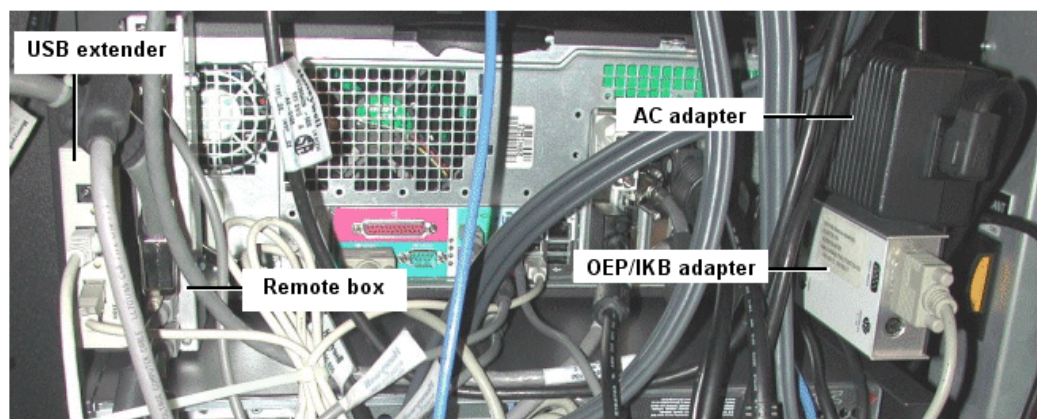


Figure 2-2 Optional Components in Cabinet

2.6 Connecting Cables to a WS360

WS360 enclosure

The following picture shows the back of the WS360 enclosure and identifies the connectors for all devices. Refer to Table 2-4 for a description of connectors. This example is configured with all possible Honeywell options.

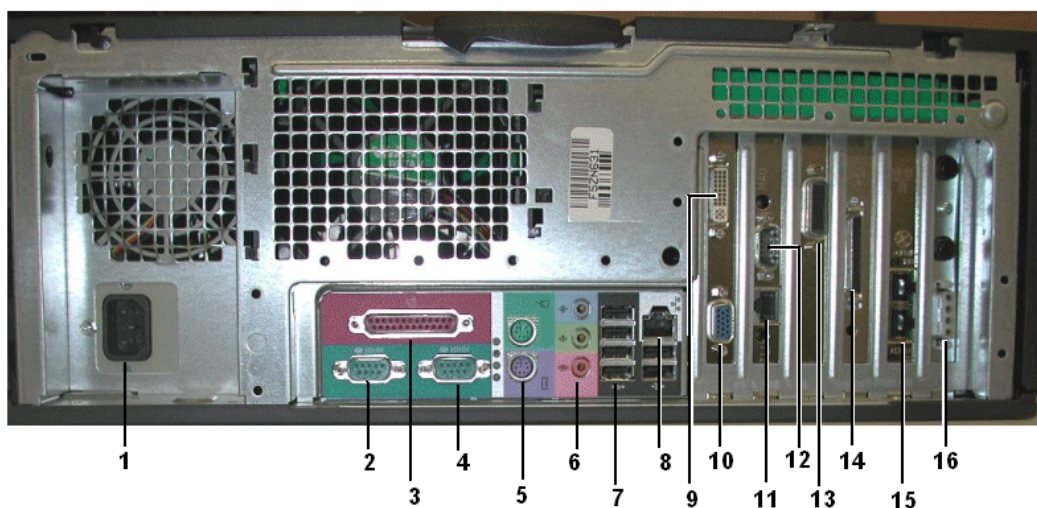


Figure 2-3 WS360 Rear View

WS360 connectors

Table 2-4 WS360 Connectors

No.	Connector	Description
1	Power cord	AC power plug
2	Serial	COM1 configured as COM3 Requires adapter for use with OEP/IKB
3	Parallel	Parallel devices such as a printer
4	Serial	COM2 configured as COM4
5	PS2	Mouse (top) Keyboard (bottom)
6	Sound	Line in connector (top)

Table 2-4 WS360 Connectors

No.	Connector	Description
		Line out connector (middle) Microphone connector (bottom)
7	USB	Use to connect USB devices installed in console
8	Network	Ethernet connection for non-FTE nodes.
9	Dual video	Use DVI to HG15 adapter if system has more than one video device
10	Analog video	Use this connector if system only has one video device
11	TLI	Use to connect systems with time synchronization
12	LCNP4	Use to connect the LCN MAU cable
13	Analog video	Use to connect a third video device, or Connect Y cable to connect a third and fourth video device
14	SCSI	Use to connect SCSI devices installed in console
15	Dual Network	Use for FTE connections
16	Remote power/circuit breaker assembly	Use for remote power

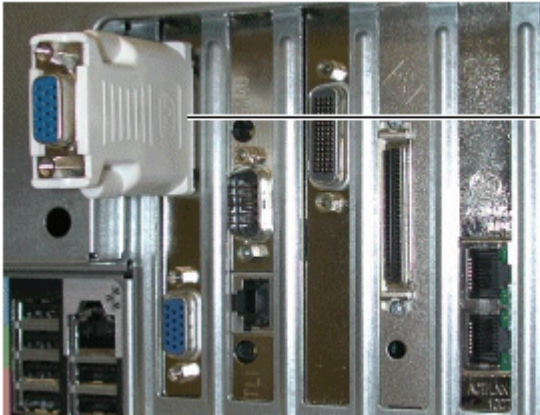
Connect cables to WS360

Use this procedure to connect the cables to the computer. Refer to Figure 2-3.

Step	Action
1	Connect the mouse and keyboard cables to the PS/2 connectors.
2	Connect the speaker cable to the audio port.
3	Connect any of the drive devices to the SCSI Controller card.
4	If you are not using Fault Tolerant Ethernet (FTE), connect the Ethernet cable to the RJ-45 connector on the Network Interface Card. If you are using FTE, you will need to connect the FTE cables according to the instructions in the <i>FTE Installation and Service Guide</i> .
5	If your console has an IKB or an OEP, connect the cables according to the instructions in Section 2.9, "Connecting Adapters".
6	Connect the LCN MAU cable to the MAU connector on the LCNP4 card if one is present.
7	Connect any USB devices or Hubs to the USB ports.

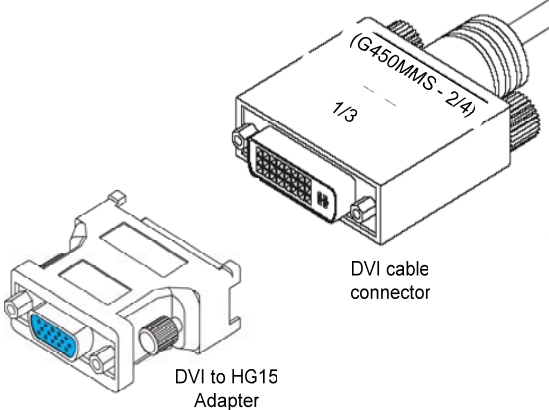
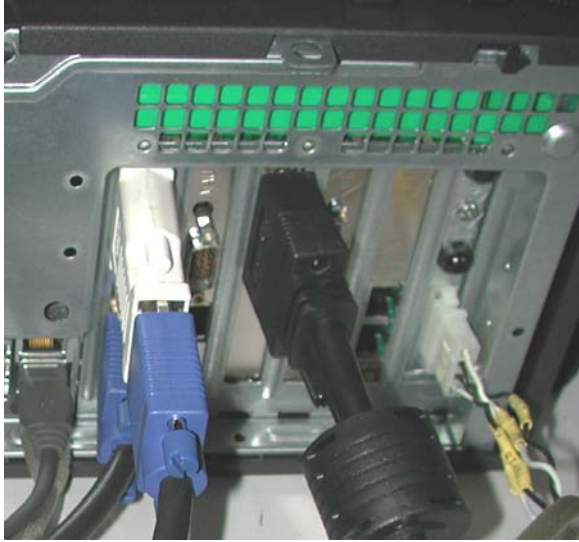
2. Platform Installation

2.6. Connecting Cables to a WS360

Step	Action
8	If your disk drive requires external power, connect the cable to the remote power.
Connecting monitors	
9	If you are connecting two or more video monitors, attach the DVI to HG15 adapter to the digital connector on the AGP card as shown.  DVI to HG15 adapter
10	Connect the primary video cable to the bottom connector. Connect the secondary video cable to the top connector (adapter).
11	If your configuration uses three or more monitors, perform the next four steps. Otherwise, secure any loose cables, and verify that all cables have proper strain relief.
12	Review the information in "Connecting monitors to the correct channel (WS360)" in this section.

2. Platform Installation

2.6. Connecting Cables to a WS360

Step	Action
13	Connect the DVI to HG15 adapters to the Y cable as shown.  DVI cable connector DVI to HG15 Adapter
14	Connect the Y cable to the video connector on the second video card (in the PCI slot). 
15	Connect the video cables according to the configuration illustrated in Figure 2-4 and Figure 2-5.
16	Secure any loose cables, and verify that all cables have proper strain relief.

2. Platform Installation

2.6. Connecting Cables to a WS360

Connecting monitors to the correct channel (WS360)

The WS360 uses a dual video card installed in the AGP slot for one or two video outputs. For systems that have three or four video outputs, the WS360 has a second dual video card installed in the appropriate PCI slot. You must connect the monitors to the correct channel identified in Table 2-5 and Figure 2-4 and Figure 2-5.

Table 2-5 WS360 Video Channels

Channel	Video Card	Four Monitor Connection	Three Monitor Connection
1	AGP video card	Bottom video card connector	Bottom video card connector
2	PCI video card	Top connector of Y cable	Top connector of Y cable
3	PCI video card	Bottom connector of Y cable	Not connected
4	AGP video card	Top video card connector	Top video card connector

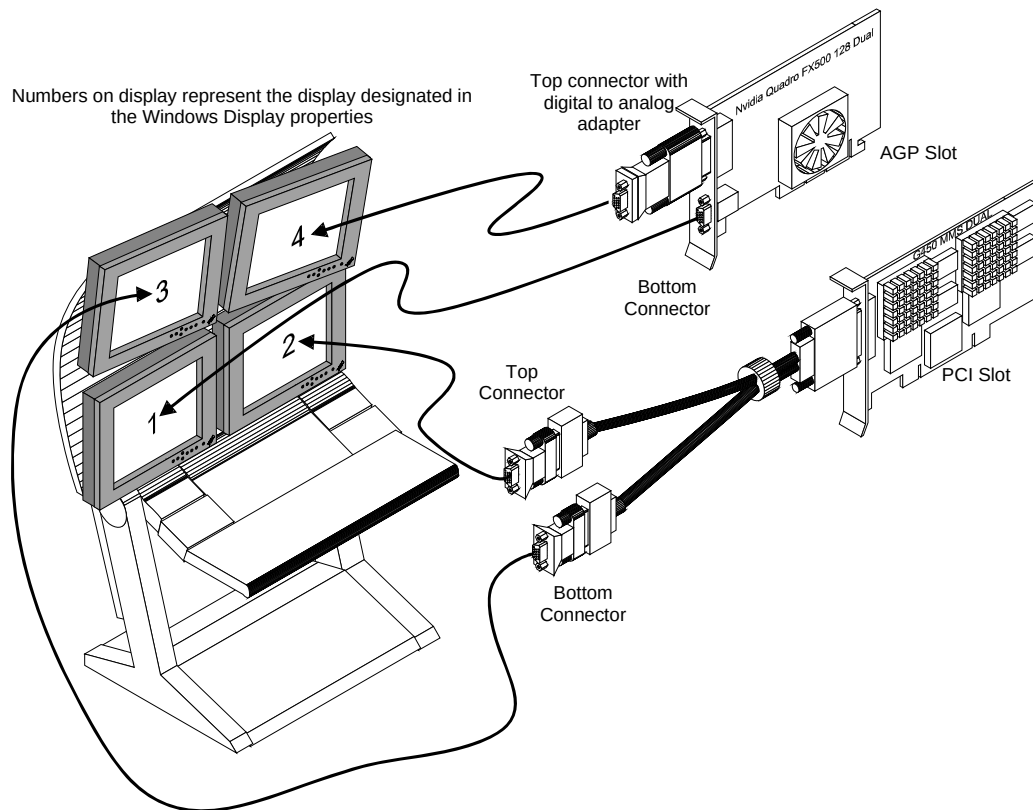


Figure 2-4 WS360 Four Monitor Configuration

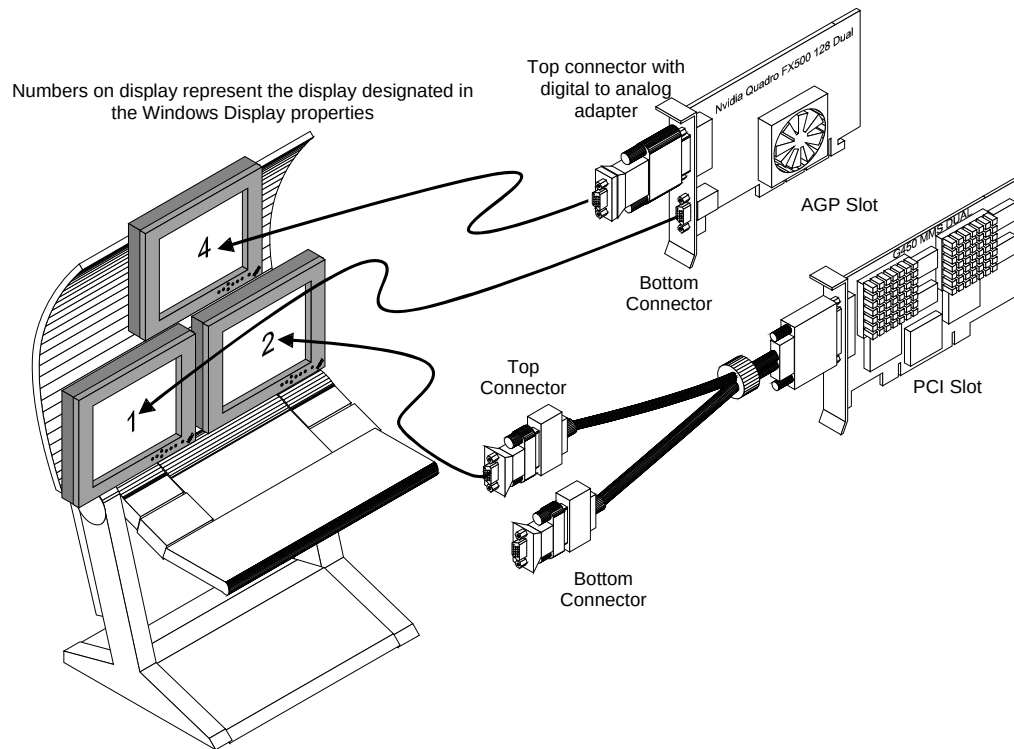


Figure 2-5 WS360 Three Monitor Configuration

2.7 Connecting Cables to a WS370

WS370 enclosure (quad video)

The following picture shows the back of the WS370 enclosure with a quad video card and additional Honeywell options. Refer to Table 2-6 for a description of connectors.

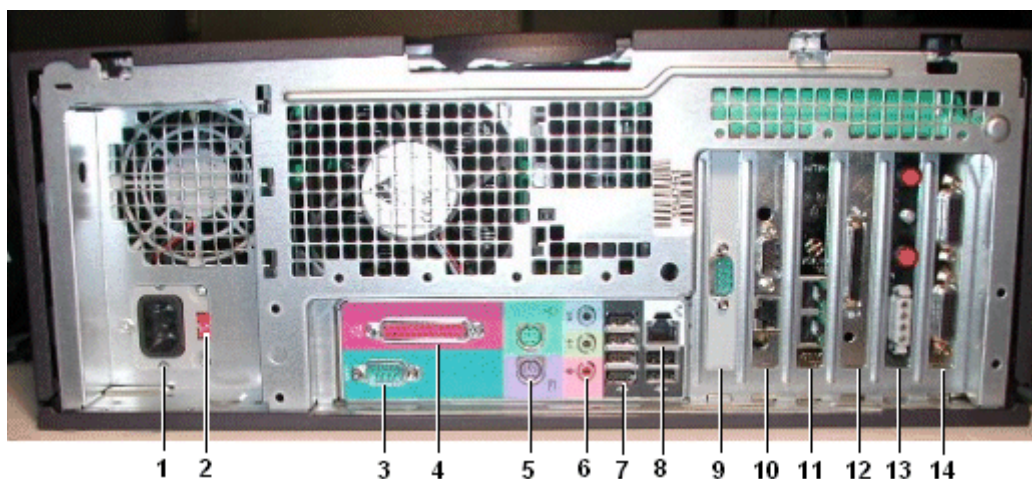


Figure 2-6 WS370 Rear View (quad video)

Table 2-6 WS370 Connectors (quad video)

No.	Connector	Description
1	Power cord	AC power plug
2	Switch	Voltage selection switch
3	Serial	COM1 configured as COM3 Requires adapter for use with OEP/IKB
4	Parallel	Parallel devices such as a printer
5	PS2	Mouse (top) Keyboard (bottom)
6	Sound	Line in connector (top) Line out connector (middle) Microphone connector (bottom)
7	USB	Use to connect USB devices installed in console

Table 2-6 WS370 Connectors (quad video)

No.	Connector	Description
8	Network	Ethernet connection for non-FTE nodes.
9	Serial	COM2 configured as COM4 Requires adapter for use with OEP/IKB
10	LCNP4	TLI (top) - Use to connect systems with time synchronization Use to connect the LCN MAU cable
11	Dual Network	Use for FTE connections
12	SCSI	Use to connect SCSI devices installed in console
13	Remote Power	Use for remote power
14	Quad video	Digital video connectors. Connect video Y cable and DVI to HG15 adapter for up to four video devices.

2. Platform Installation

2.7. Connecting Cables to a WS370

WS370 enclosure (single/dual video)

The following picture shows the back of the WS370 enclosure with a dual video card and additional Honeywell options. Refer to for a description of connectors.

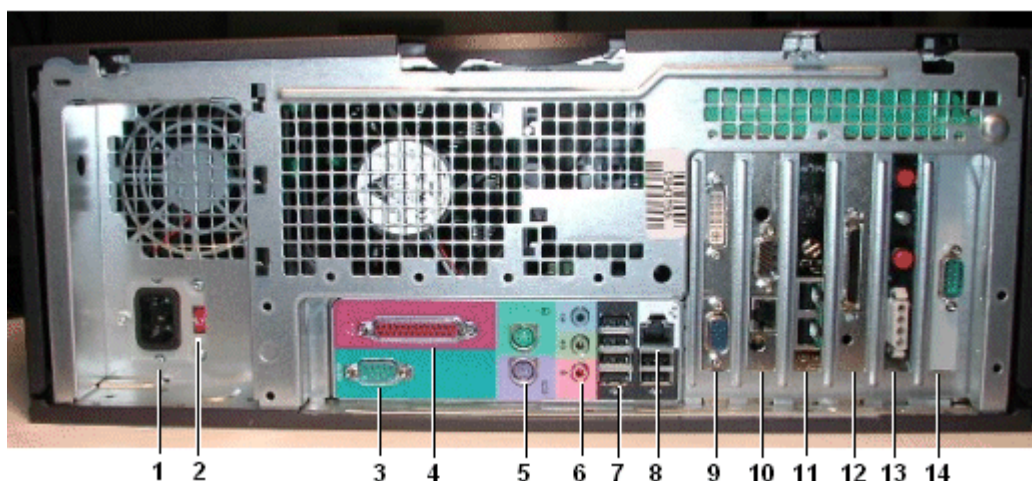


Figure 2-7 WS370 Rear View (dual video)

Table 2-7 WS370 Connectors (dual video)

No.	Connector	Description
1	Power cord	AC power plug
2	Switch	Voltage selection switch
3	Parallel	Parallel devices such as a printer
4	Serial	COM1 configured as COM3 Requires adapter for use with OEP/IKB
5	PS2	Mouse (top) Keyboard (bottom)
6	Sound	Line in connector (top) Line out connector (middle) Microphone connector (bottom)
7	USB	Use to connect USB devices installed in console
8	Network	Ethernet connection for non-FTE nodes.

Table 2-7 WS370 Connectors (dual video)

No.	Connector	Description
9	Dual video	Use DVI to HG15 adapter if system has more than one video device.
10	LCNP	LCN (top) Use to connect the LCN MAU cable TLI (bottom) Use to connect systems with time synchronization
11	Dual Network	Use for FTE connections
12	SCSI	Use to connect SCSI devices installed in console
13	Remote Power	Use for remote power
14	Serial	COM2 configured as COM4

Connect cables to WS370

Use this procedure to connect the cables to the computer. Refer also to Figure 2-6.



CAUTION

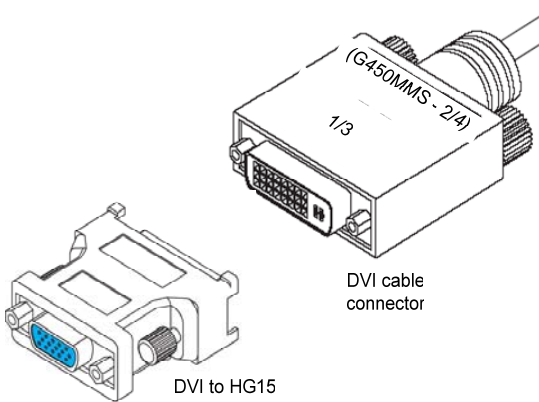
The WS370 platform has a voltage selection switch next to the AC power cord connector.

Before connecting power, verify the switch is in the correct position for the type of voltage you are using.

Step	Action
1	Connect the mouse and keyboard cables to the PS/2 connectors.
2	Connect the speaker cable to the audio port.
3	Connect any of the drive devices to the SCSI Controller card.
4	If you are not using Fault Tolerant Ethernet (FTE), connect the Ethernet cable to the RJ-45 connector on the Network Interface Card. If you are using FTE, you will need to connect the FTE cables according to the instructions in the <i>FTE Installation and Service Guide</i> .
5	If your console has an IKB or an OEP, connect the cables according to the instructions in Section 2.9, "Connecting Adapters".
6	Connect the LCN MAU cable to the MAU connector on the LCNP4 card if one is present.
7	Connect any USB devices or Hubs to the USB ports.
8	If your disk drive requires external power, connect the cable to the remote power.

2. Platform Installation

2.7. Connecting Cables to a WS370

Step	Action
Connecting monitors (dual video card)	
9	If you are connecting two video devices, connect the DVI to HG15 adapter to the top digital connector. 
10	Connect the primary video cable to the bottom connector.
11	Connect the secondary video cable to the top converter.
Connecting monitors (quad video card)	
9	Connect the DVI to HG15 adapters to the Y cable as shown.  DVI cable connector DVI to HG15 Adapter
10	Connect the Y cables to the video connectors on the video card.
11	Connect the video cables according to the configuration listed in Table 2-8
12	Secure any loose cables, and verify that all cables have proper strain relief.

Connecting monitors to the correct channel (WS370)

If you ordered the quad video option, the video card will be installed in the PCI4 slot. For systems that have three or four video outputs, the WS370 uses two video Y cables connected to the quad video card. You must connect the monitors to the correct channels identified in Table 2-8. See also Figure 2-8 and Figure 2-9.

Table 2-8 WS370 Video Channels

Channel	Four Monitor Connection	Three Monitor Connection
1	Top connector of upper Y cable	Top connector of upper Y cable
2	Bottom connector of upper Y cable	Bottom connector of upper Y cable
3	Top connector of lower Y cable	Top connector of lower Y cable
4	Bottom connector of lower Y cable	Not used

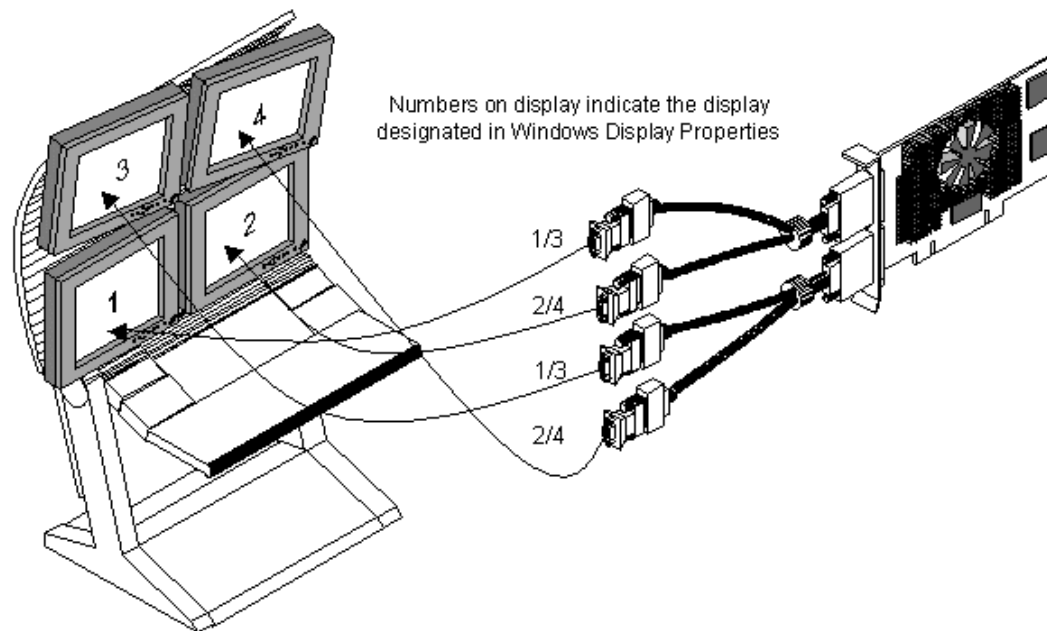


Figure 2-8 WS370 Four Monitor Configuration

2. Platform Installation

2.7. Connecting Cables to a WS370

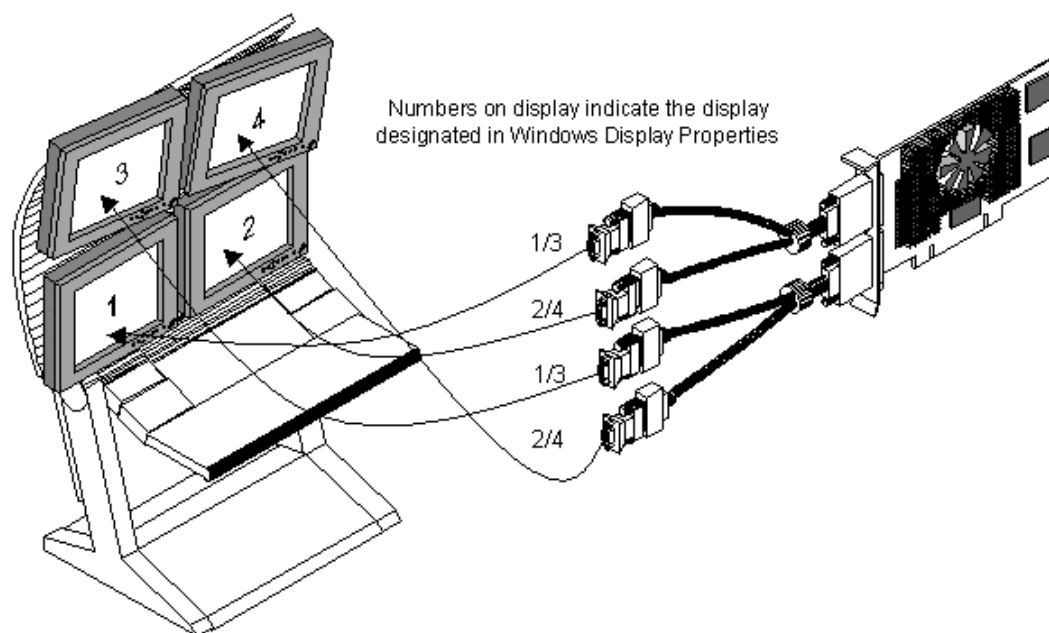


Figure 2-9 **WS370 Three Monitor Configuration**

2.8 Connecting Cables to a WS470

WS470 enclosure

This section contains information for connecting cables, including monitor(s), to the WS470 platform.



Figure 2-10 WS470 Rear View

Standard WS470 connections

The following figure shows the standard connections for a WS470 platform. Refer to Table 2-9 for a description of connections.

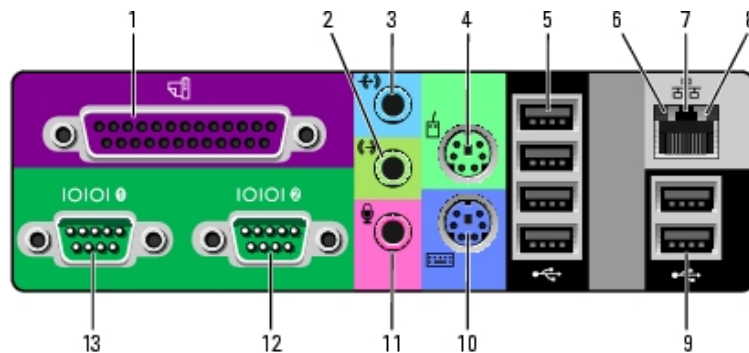


Figure 2-11 WS470 standard connections

2. Platform Installation

2.8. Connecting Cables to a WS470

Table 2-9 WS470 standard connections

No.	Connector	Description
1	parallel connector	Connect a parallel device, such as a printer, to the parallel connector. If you have a USB printer, plug it into a USB connector. NOTE: The integrated parallel connector is automatically disabled if the computer detects an installed card containing a parallel connector configured to the same address. For more information, see "System Setup Options."
2	line-in connector	Use the blue line-in connector to attach a record/playback device such as a cassette player, CD player, or VCR. On computers with a sound card, use the connector on the card.
3	line-out port	Use the green line-out connector to attach headphones and most speakers with integrated amplifiers. On computers with a sound card, use the connector on the card.
4	mouse port	Plug a standard mouse into the green mouse connector. Turn off the computer and any attached devices before you connect a mouse to the computer. If you have a USB mouse, plug it into a USB connector. If your computer is running the Microsoft® Windows® XP operating system, the necessary mouse drivers have been installed on your hard drive.
5	USB 2.0 ports (4)	Use the back USB connectors for devices that typically remain connected, such as printers and keyboards. It is recommended that you use the front USB connectors for devices that you connect occasionally, such as flash memory key or cameras, or for bootable USB devices.

Table 2-9 WS470 standard connections

No.	Connector	Description
6	link integrity light	Green — A good connection exists between a 10-Mbps network and the computer. Orange — A good connection exists between a 100-Mbps network and the computer. Yellow — A good connection exists between a 1000-Mbps (or 1-Gbps) network and the computer. Off — The computer is not detecting a physical connection to the network.
7	network adapter port	To attach your computer to a network or broadband device, connect one end of a network cable to either a network jack or your network or broadband device. Connect the other end of the network cable to the network adapter connector on your computer. A click indicates that the network cable has been securely attached. NOTE: Do not plug a telephone cable into the network connector. On computers with a network connector card, use the connector on the card. It is recommended that you use Category 5 wiring and connectors for your network. If you must use Category 3 wiring, force the network speed to 10 Mbps to ensure reliable operation.
8	network activity light	Flashes a yellow light when the computer is transmitting or receiving network data. A high volume of network traffic may make this light appear to be in a steady "on" state.
9	USB 2.0 ports (2)	Use the back USB connectors for devices that typically remain connected, such as printers and keyboards. It is recommended that you use the front USB connectors for devices that you connect occasionally, such as flash memory key or cameras, or for bootable USB devices. A USB connector is also used for connecting a touchscreen to a flat panel display on an Icon console.

2. Platform Installation

2.8. Connecting Cables to a WS470

Table 2-9 WS470 standard connections

No.	Connector	Description
10	keyboard port	If you have a standard keyboard, plug it into the purple keyboard connector. If you have a USB keyboard, plug it into a USB connector.
11	microphone port	Use the pink microphone connector to attach a personal computer microphone for voice or musical input into a sound or telephony program. On computers with a sound card, the microphone connector is on the card.
12	Serial port	COM2 configured as COM4 Used to connect a serial device or an OEP/IKB adapter.
13	Serial port	COM1 configured as COM3 Used to connect a serial device or a Honeywell touchscreen adapter.

WS470 connections for Honeywell installed components

The following figure shows the connections for the Honeywell installed components on the WS470 platform. Refer to Table 2-10 for a description of connections.

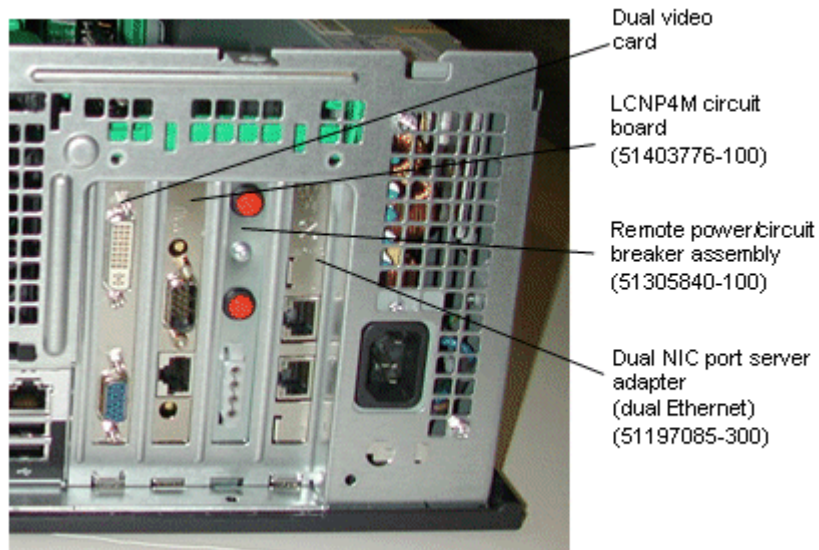


Figure 2-12 WS470 Honeywell installed components

Table 2-10 WS470 connections for Honeywell installed components

No	Slot	Description
1	PClex x16	Dual video card (one or two video configuration)
2	PCI	LCNP4M board
3	PCIe x 4	Remote power/circuit breaker assembly
4	PCIX	Dual NIC

2. Platform Installation

2.8. Connecting Cables to a WS470

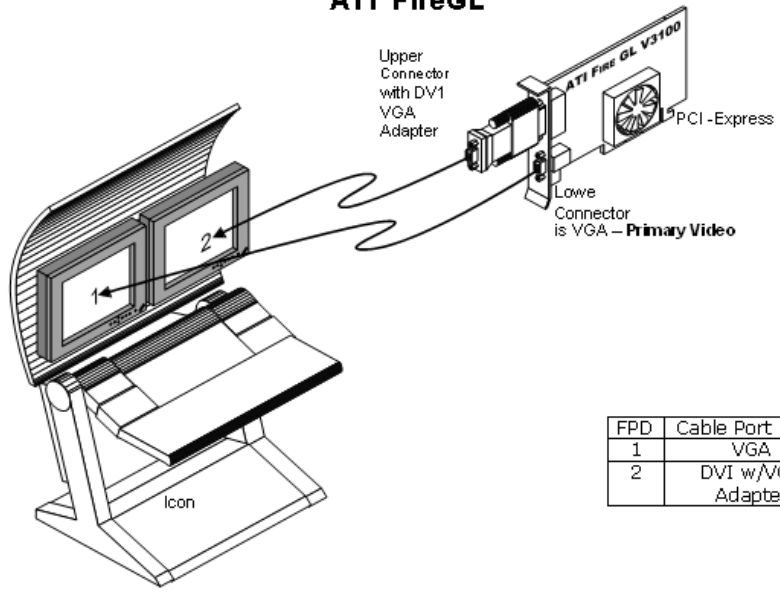
Connect remaining WS470 cables and power

Use this procedure to connect the cables to the WS470 computer. Refer also to Figure 2-11 and Figure 2-12.

Step	Action
1	Connect the mouse cables to the PS/2 port.
2	If you have an IKB that uses the OEP/IKB adapter: • Connect the PS/2 keyboard cable from the OEP/IKB adapter to the keyboard PS/2 port. • Connect the DB9 serial extension cable to the serial port.
3	If you are not using Fault Tolerant Ethernet (FTE), connect the Ethernet cable to the RJ-45 connector on the Network Interface Card. If you are using FTE, you will need to connect the FTE cables according to the instructions in the <i>FTE Installation and Service Guide</i> .
4	Connect the LCN MAU cable to the MAU connector on the LCNP4M card.
5	Connect any USB devices or Hubs to the USB ports, including the USB IKB if you are using one.

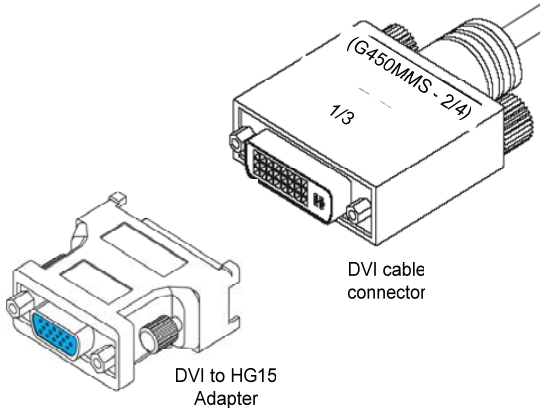
2. Platform Installation

2.8. Connecting Cables to a WS470

Step	Action						
Connecting monitors (single/dual video card)							
6	If you are connecting two video devices, connect the DVI to HG15 adapter to the top digital connector.						
ATI FireGL  <table border="1" data-bbox="1068 1031 1299 1123"> <thead> <tr> <th>FPD</th><th>Cable Port</th></tr> </thead> <tbody> <tr> <td>1</td><td>VGA</td></tr> <tr> <td>2</td><td>DVI w/VGA Adapter</td></tr> </tbody> </table> VideoATI		FPD	Cable Port	1	VGA	2	DVI w/VGA Adapter
FPD	Cable Port						
1	VGA						
2	DVI w/VGA Adapter						
7	Connect the primary video cable to the bottom connector.						
8	Connect the secondary video cable to the top converter.						

2. Platform Installation

2.8. Connecting Cables to a WS470

Step	Action
Connecting monitors (quad video card)	
9	Connect the DVI to HG15 adapters to the Y cable as shown. 
10	Connect the Y cables to the video connectors on the video card.
11	Connect the video cables according to the information in the following "Connect Icon monitors" procedure.
12	Secure any loose cables, and verify that all cables have proper strain relief.
13	You are now ready to start and configure your node. Go to the Section 3, "Computer Operation".

Connect Icon monitors

For systems that have three or four video outputs, the WS470 uses two video Y cables connected to a quad video card. You must connect the monitors to the correct channels identified in Table 2-8. See also Figure 2-8 and Figure 2-9.

Table 2-11 WS470 Video Channels

Channel	Four Monitor Connection	Three Monitor Connection
1	Top connector of upper Y cable	Top connector of upper Y cable
2	Bottom connector of upper Y cable	Bottom connector of upper Y cable
3	Top connector of lower Y cable	Top connector of lower Y cable
4	Bottom connector of lower Y cable	Not used

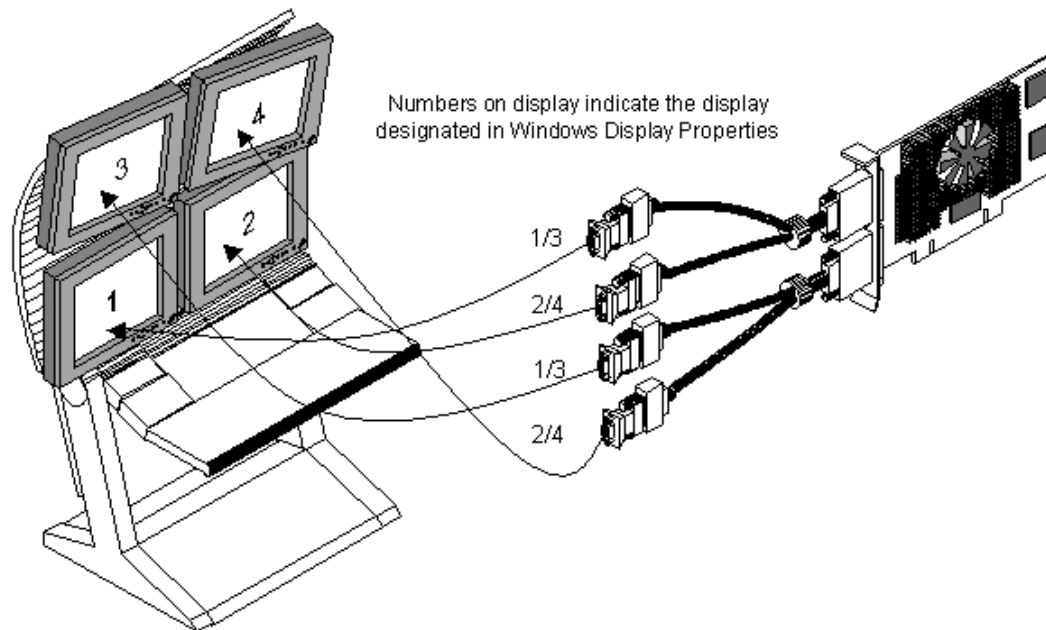


Figure 2-13 WS470 Four Monitor Configuration

2. Platform Installation

2.8. Connecting Cables to a WS470

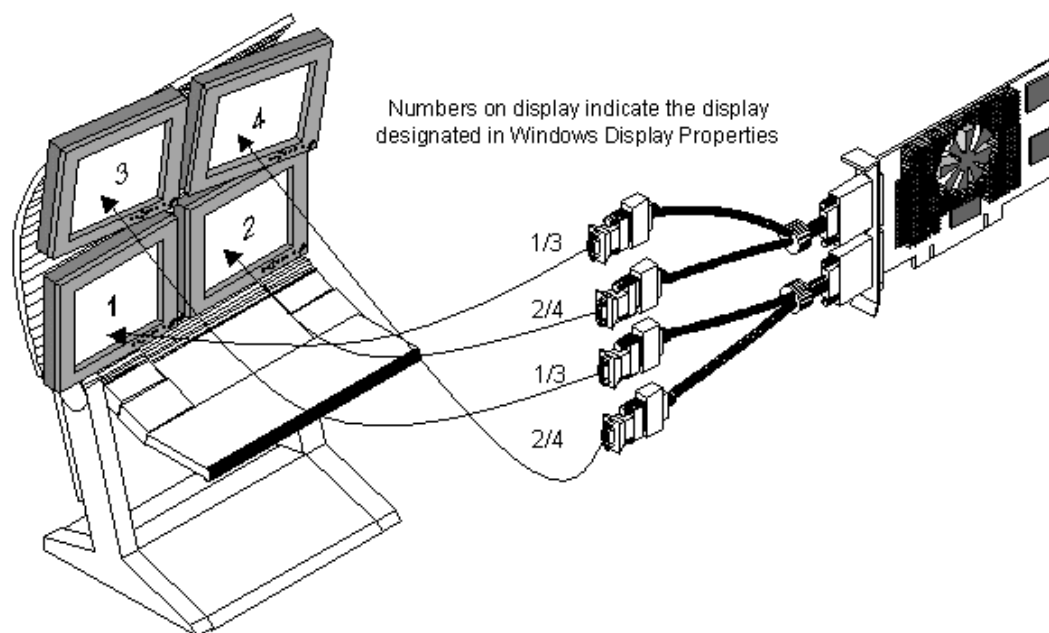


Figure 2-14 WS470 Three Monitor Configuration

2.9 Connecting Adapters

WS360 serial ports

The WS360 has two standard serial ports: COM 1 and COM 2. To ensure these ports work correctly with Honeywell hardware, the serial ports are reconfigured as COM 3 and COM 4 respectively when the computer is initialized. See Figure 2-3 to view the serial ports. You must connect Honeywell devices to the correct serial port as defined in the following table.

Table 2-12 WS360 COM Port Connections

Honeywell Device	Connect to this Serial Port
OEP	COM1 (configured as COM3)
IKB	COM1 (configured as COM3)
Non-USB Touchscreen	COM2 (configured as COM4)

WS370 serial ports

The WS370 has one standard serial port and an additional serial port installed in the PCI4 or the PCI Express slot, depending on your configuration. To ensure these ports work correctly with Honeywell hardware, the WS370 serial ports are reconfigured as COM 3 and COM 4 respectively when the computer is initialized. See Figure 2-6 and Figure 2-7 to view the serial ports. You must connect Honeywell devices to the correct serial port as defined in the following table.

Table 2-13 WS370 COM Port Connections

Honeywell Device	Connect to this Serial Port
OEP	COM1 (configured as COM3)
IKB	COM1 (configured as COM3)
Non-USB Touchscreen	COM2 (configured as COM4)

WS470 serial ports

The WS470 has two standard 9-pin serial ports COM 1 and COM 2. To ensure these ports work correctly with Honeywell hardware, the WS470 serial ports are reconfigured as COM 3 and COM 4 respectively when the computer is initialized. See Figure 2-10 and Figure 2-11 (call out number 12 and 13) to view the serial ports. You must connect Honeywell devices to the correct serial port as defined in the following table.

Table 2-14 WS470 COM Port Connections

Honeywell Device	Connect to this Serial Port
OEP	COM1 (configured as COM3)
IKB	COM1 (configured as COM3)
Non-USB Touchscreen	COM2 (configured as COM4) Note: A USB port is used for a flat panel display touchscreen on the Icon console.

OEP/IKB Adapter Configurations

Purpose

Use the information and procedures in this section to connect the OEP/IKB adapter. The following block diagrams show the basic connections for the OEP/IKB adapter assembly used in different configurations.

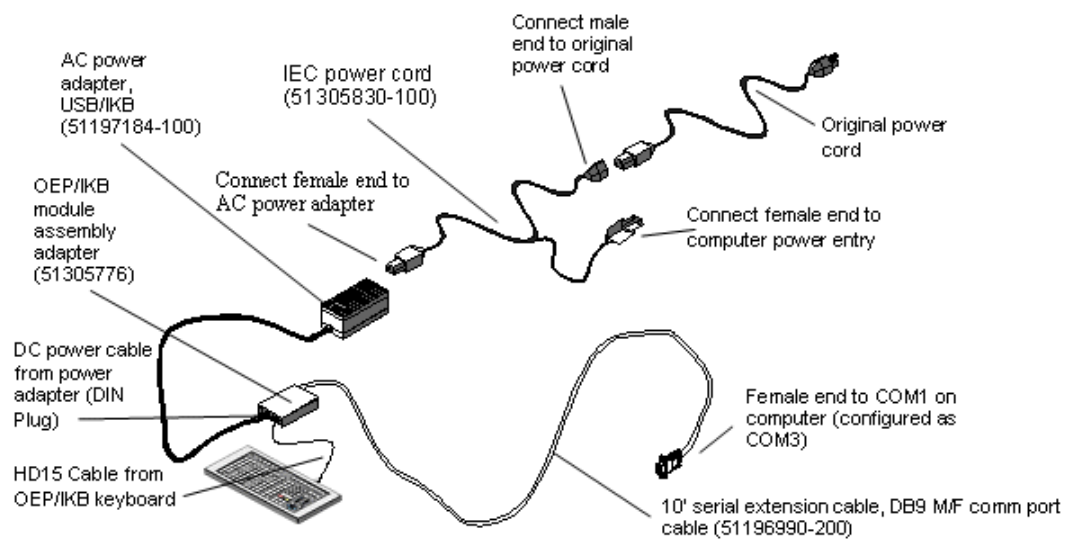


Figure 2-15 OEP/IKB Adapter Connections for OEP

2. Platform Installation

2.9. Connecting Adapters

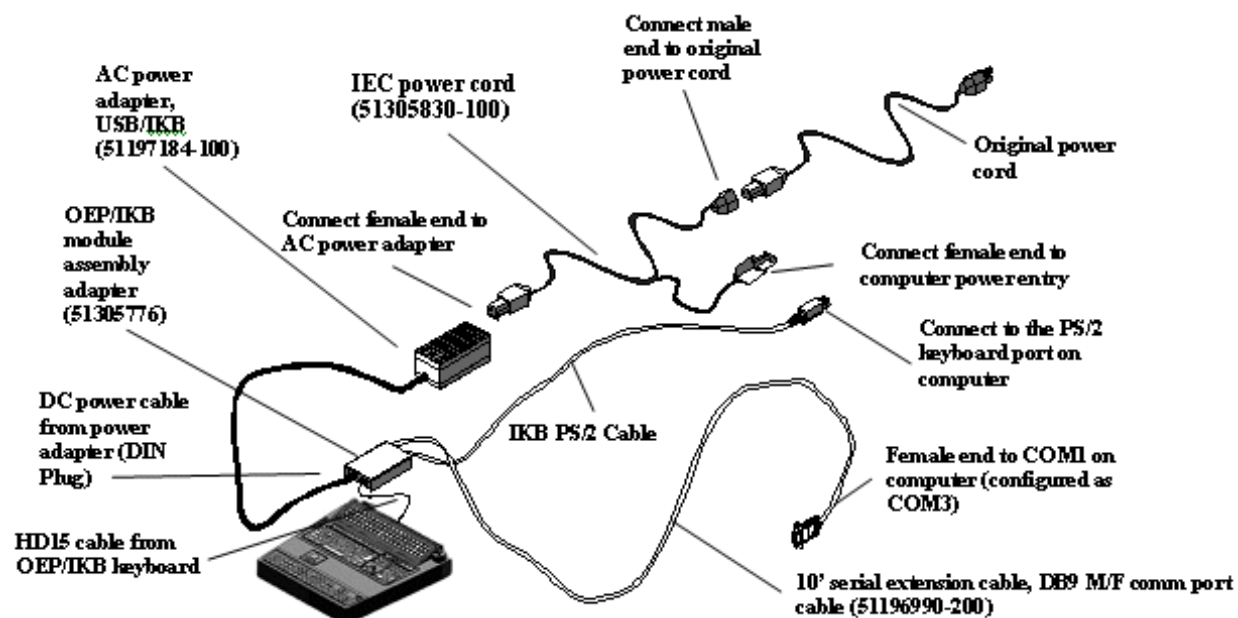


Figure 2-16 OEP/IKB Adapter Connections for IKB

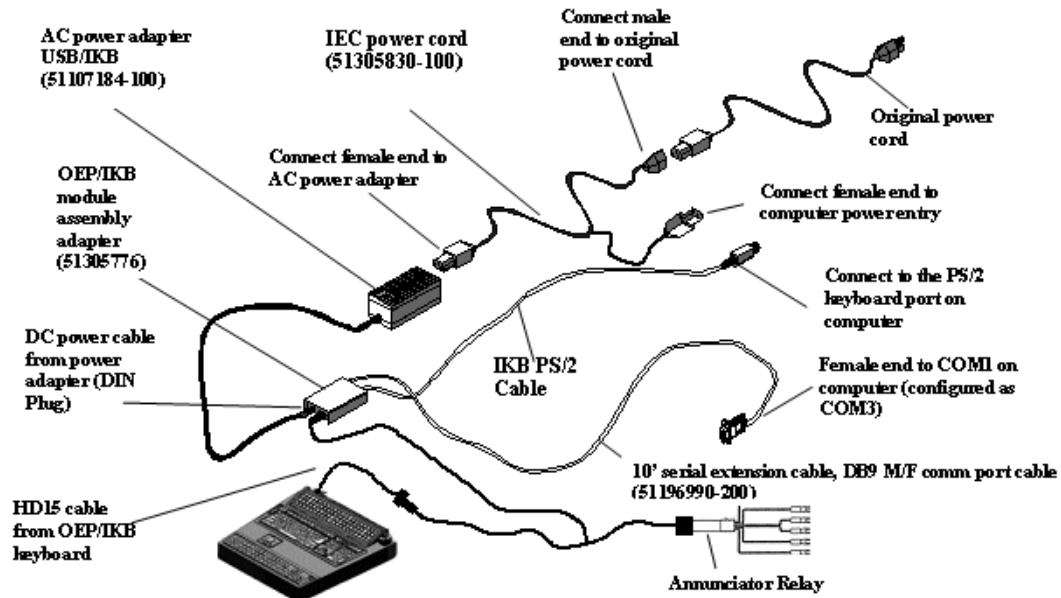


Figure 2-17 OEP/IKB Adapter Connections for IKB with Annunciator Relay

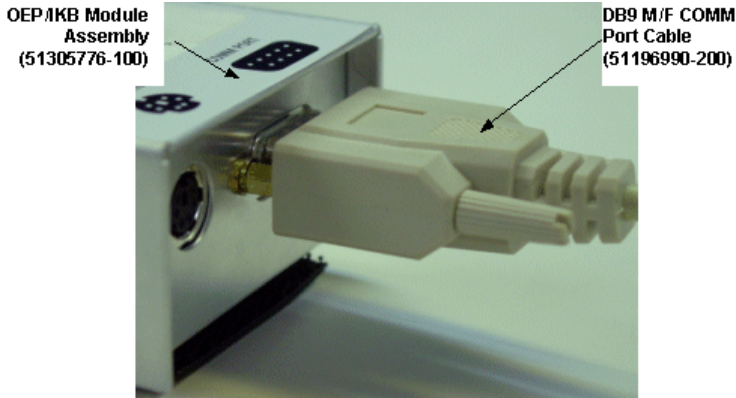
Install OEP/IKB adapter

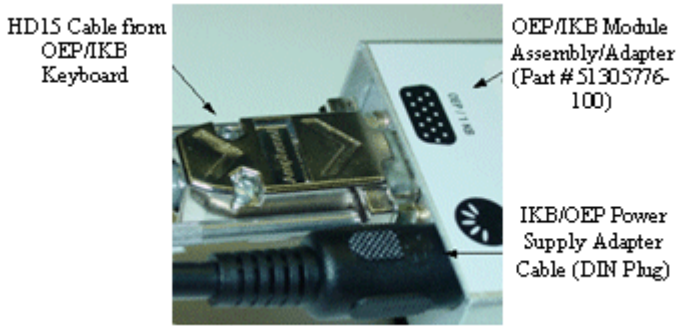
Use this procedure to connect the OEP/IKB adapter.

Step	Action
1	Locate a safe location for the OEP/IKB adapter box and secure it using the supplied Velcro.
2	Locate a safe location to place the OEP/IKB power adapter box and secure it using the supplied Velcro.
3	If The IKB has a trackball connect it to the mouse PS/2 port on the PC.

2. Platform Installation

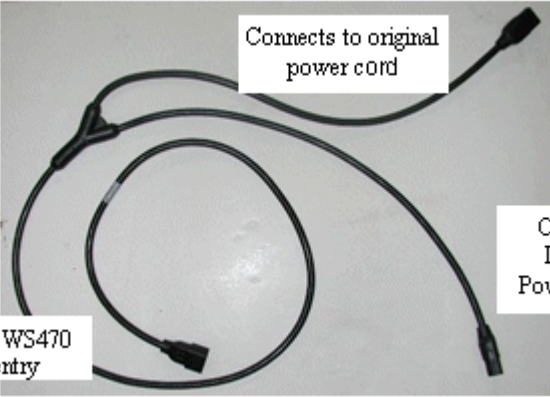
2.9. Connecting Adapters

Step	Action
4	Locate and connect the male end of the DB9 M/F COM Port Cable (51196990-200) to the COM PORT on the OEP/IKB Module Assembly/Adapter (51305776-100).  The image shows a close-up of the OEP/IKB Module Assembly (51305776-100) with a DB9 M/F COM Port Cable (51196990-200) plugged into its COM PORT. Arrows point from the labels to the respective components.
5	Secure the connection by tightening the thumbscrews located on the cable end.
6	Connect the female end of the DB9 M/F COM Port Cable (51196990-200) to COM 1 port located on the computer's back panel and tighten the thumbscrews.
7	Insert the round DIN plug of the supplied IKB PS/2 cable (51305381-300) into the PS/2 Connector on the OEP/IKB Module Assembly (51305776-100).  The image shows the OEP/IKB Module Assembly (51305776-100) with the IKB PS/2 Cable (51305381-300) plugged into its PS/2 Connector. A DB9 M/F COM Port Cable (51196990-200) is also visible. Arrows point from the labels to the respective components.
8	Connect the free end of the IKB PS/2 cable (51305381-300) to the Keyboard PS/2 Port on the computer.

Step	Action
9	Take the DC power cable from the IKB/OEP Power Supply Adapter (51197184-100) and insert the round DIN plug into the Power Connector receptacle on the IKB/OEP Module Assembly/Adapter (51305776-100).  The diagram shows a close-up of the connection. A black cable with a round DIN plug is being inserted into a receptacle on a white module assembly. Labels point to the 'HD15 OEP/IKB Receptacle', 'OEP/IKB Module Assembly/Adapter (Part # 51305776-100)', 'Power Connector Receptacle', 'HD15 Cable from OEP/IKB', and 'AC Power Source DC Power Cable (51197184-100)'.
10	Connect the existing OEP/IKB Keyboard HD15 Cable to the HD15 OEP/IKB receptacle located on the IKB/OEP Assembly/Adaptor (51305776-100). Secure cable to the assembly/adaptor by tightening the thumbscrews located on the cable's end.  The diagram shows a close-up of the connection. A keyboard cable with a DIN plug is being inserted into a receptacle on a white module assembly. Labels point to the 'HD15 Cable from OEP/IKB Keyboard', 'OEP/IKB Module Assembly/Adapter (Part # 51305776-100)', and 'IKB/OEP Power Supply Adapter Cable (DIN Plug)'.

2. Platform Installation

2.9. Connecting Adapters

Step	Action
11	Connect the IEC power cord (51305830-100): • Connect one female end to the OEP/IKB power adapter (51197184-100). • Connect the other female end to the WS470 power entry. • Connect the male end to the original computer power cord. 

Connect Touchscreen Power Adapter Cable

Purpose

Use the information and procedures in this section to connect the touchscreen power cable for the following touchscreen configurations:

- Touchscreen is connected to the touchscreen adapter box.
- Touchscreen is connected to the DB9 socket on the back of the monitor.
- Touchscreen is connected to the DB9 socket on the touchscreen frame.

Connect touchscreen adapter cable

Use this procedure to connect the supplied touchscreen adapter cable to the touchscreen device and the computer.

Step	Action
1	Remove the existing DB9 touchscreen cable.
2	If your touchscreen uses the touchscreen adapter box, connect the end of the new touchscreen cable that does not have the 4-pin connector to the DB9 socket on the box as shown. 
3	If your touchscreen uses the DB9 socket on the back of the monitor, connect the end of the new touchscreen cable that does not have the 4-pin connector to the DB9 socket on the back of the monitor.
4	If your touchscreen uses the DB9 socket on the touchscreen frame, connect the end of the new touchscreen cable that does not have the 4-pin connector to the DB9 socket on the touchscreen frame. Notes: If 3 or 4 do not apply to your configuration you must upgrade to a flat panel display for touchscreen functionality. If you are mounting in a console the CRT cover needs to be removed.
5	Connect the other end of the touchscreen cable (end with the 4-pin connector) to the COM2 serial port on the computer.
6	Connect the 4-pin connector to the remote power circuit assembly card on the computer.

2. Platform Installation

2.9. Connecting Adapters

3. Computer Operation

3.1 Starting up your Computer

Overview

The following table lists those tasks that must be performed before operating your computer. This section contains step-by-step procedures for each of these tasks.

✓	Task
	Turn on power and check status
	Check LCNP status
	Set monitor resolution
	Configure three monitors for Icon Consoles

Turn on power and check status

Step	Action
1	Press the power button on the front panel of the computer.
2	Wait for the power light to become solid green.
3	If the power light does not become solid green, refer to the "Advanced Troubleshooting" section of the <i>Dell Precision™ Workstation 360 User's Guide</i> .

3. Computer Operation

3.1. Starting up your Computer

Check LCNP status

If this is a TPS node with the LCNP4 board installed, verify that the LCNP passed self-test.

Step	Action
1	From the Start menu, select LCNP Status .
2	Verify that the LCNP status indicates Passed Self Test and the circle is green.
3	Verify that LCN Address appears in the LEDs field of the LCNP Status display.



REFERENCE

Refer to the LCNP Status section in the *LCNP Status User's Guide* for more information.

Set monitor resolution

When the computer is initialized, the monitor configuration is established based on the following user input:

- CRT type monitors: 75 Hz
- FPD type monitors: 60 Hz

If, after initialization, you install a monitor type different than what was originally defined, you will need to adjust the monitor settings for optimal performance. See your specific monitor user guide for other recommended settings.

Step	Action
1	Select Start > Settings > Control Panel .
2	Double-click the Display icon and select the Settings tab.
3	Under Screen area, drag the slider to the appropriate resolution.
4	Click Apply .

Configure three monitors for Icon Consoles

Use this procedure if you have a quad video configuration and are not using all four of the available video display connectors on the installed video cards.



ATTENTION

If you are not using all available channels on the installed video cards, you must configure your monitors such that the monitors are connected to the lowest and highest channels.

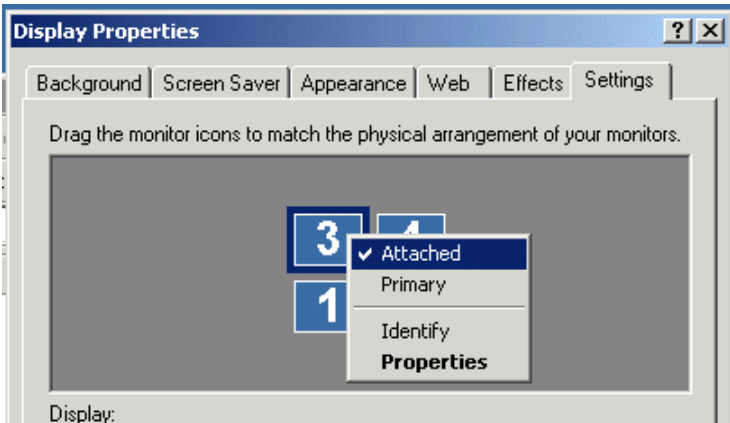
For example: If you are using 3 monitors with the two dual video cards installed, you must:

- Attach the monitors to channels 1, 2 and 4.
- In the monitor settings, disable monitor 3 and leave monitors 1, 2 and 4 attached.

Step	Action
1	Verify that the monitors are physically attached to at least the highest and lowest video channels. See • Figure 2-5 WS360 Three Monitor Configuration, or • Figure 2-9 WS370 Three Monitor Configuration • Figure 2-14 WS470 Three Monitor Configuration
2	Select Start > Settings > Control Panel .
3	Double-click the Display icon and select the Settings tab.
4	Right-click on monitor to be disabled (most likely number 3).

3. Computer Operation

3.1. Starting up your Computer

Step	Action
5	Select Attached to uncheck the option and disable the monitor. Click OK. 

3.2 WS360 Overview

Front view of WS360 enclosure

The following figure shows the front view of the WS360 enclosure.

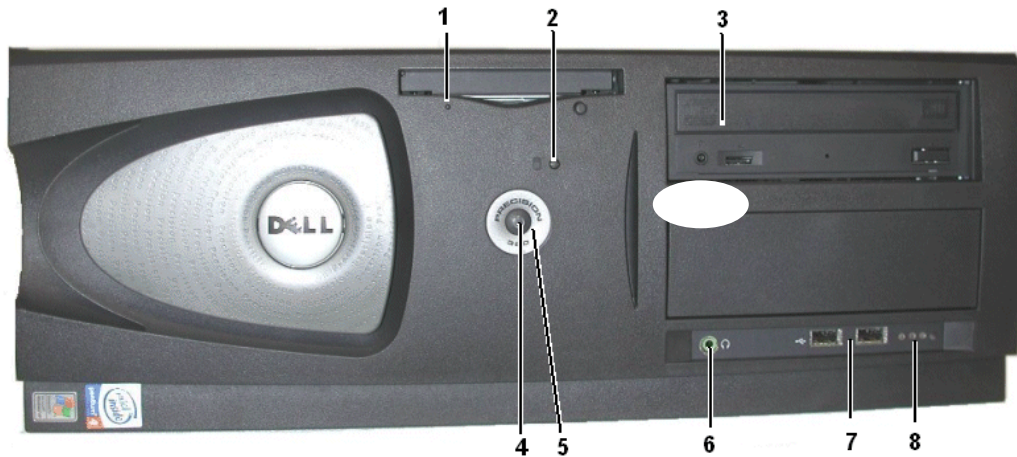


Figure 3-1 Front View of WS360 Enclosure

Table 3-1 WS360 Front Panel

No.	Control	Description
1	Floppy-drive light button	The floppy-drive light is on when the computer reads data from or writes data to the floppy drive. Wait until this light turns off before you remove the floppy disk from the drive.
2	Hard-drive light	The hard-drive light is on when the computer reads data from or writes data to the hard drive. The light might also be on when a device such as your CD player is operating.
3	CD/DVD drive	Access the CD or DVD drive.

3. Computer Operation

3.2. WS360 Overview

Table 3-1 WS360 Front Panel

No.	Control	Description
4	Power light	The power light illuminates and blinks or remains solid to indicate the following different states: No light — The computer is in the off state. Steady green —The computer is in a normal operating state. Blinking green —The computer is in a power-saving state. To exit from a power-saving state, press the power button or use the keyboard or the mouse if it is configured as a wake device in the Windows Device Manager. For more information about sleep states and exiting from a power-saving state, see "Power Button" in the <i>Dell Precision™ Workstation 360 User's Guide</i>. See "Diagnostic Lights" in the <i>Dell Precision™ Workstation 360 User's Guide</i> for a description of light codes that can help you troubleshoot problems with your computer.
5	Power button	Press this button to turn on the computer.
6	Headphone connector	Attach headphones.
7	USB 2.0 connectors (2)	Use the front USB connectors for devices that you connect occasionally, such as joysticks or cameras. It is recommended that you use the back USB connectors for devices that typically remain connected, such as printers and keyboards.
8	Diagnostic lights (4)	Use the lights to help you troubleshoot a computer problem based on the diagnostic code. For more information, see "Diagnostic Lights" in the <i>Dell Precision™ Workstation 360 User's Guide</i> .

Inside WS360 enclosure

The following picture shows the inside of the WS360 enclosure and identifies the areas within the enclosure.

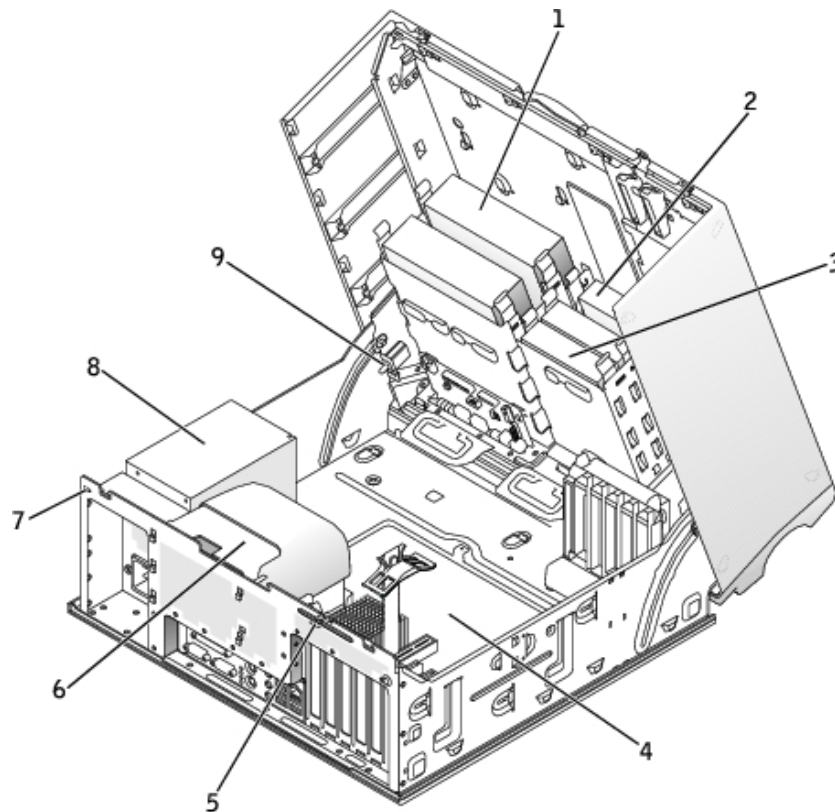


Figure 3-2 Inside WS360 Enclosure

- | | | | |
|----------|--------------|----------|-------------------------------|
| 1 | CD/DVD drive | 6 | Microprocessor airflow shroud |
| 2 | Floppy drive | 7 | Security cable slot |
| 3 | Hard drive | 8 | Power supply |
| 4 | System board | 9 | Chassis intrusion switch |
| 5 | Padlock ring | | |

3. Computer Operation

3.2. WS360 Overview

Honeywell options for the WS360

The following picture shows the inside of the WS360 enclosure and identifies the options installed by Honeywell.

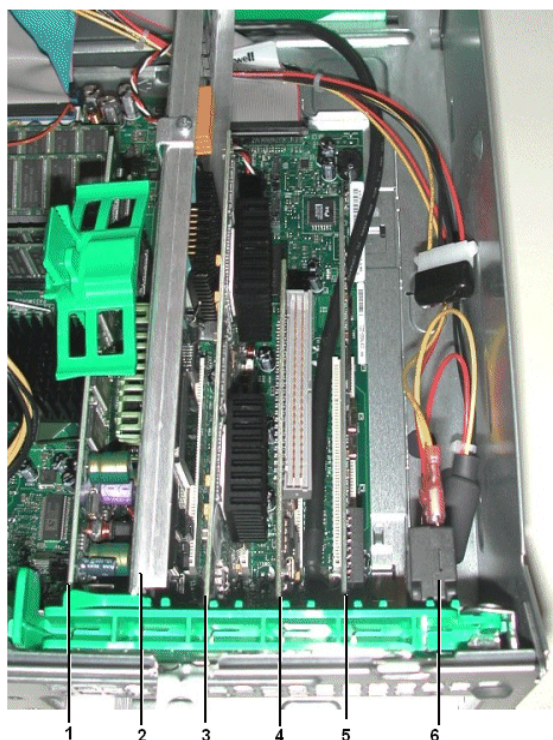


Figure 3-3 Honeywell Options

No.	Slot	Description of Card
1	AGP	Dual head AGP video card
2	PCI1	LCNP4 board
3	PCI2	Dual video card
4	PCI3	SCSI controller card
5	PCI4	Intel dual NIC for FTE
6	Empty	Remote power connection

Device cable Colors

The devices within the enclosure have specific colors as listed in the following table.

Device	Cable Color
Hard drive	Blue pull tab
Floppy drive	Black pull tab
CD/DVD drive	Orange pull tab
I/O panel	Yellow pull tab

3.3 WS370 Overview

Front view of enclosure (WS370)

The following figure shows the front view of the WS370 enclosure.

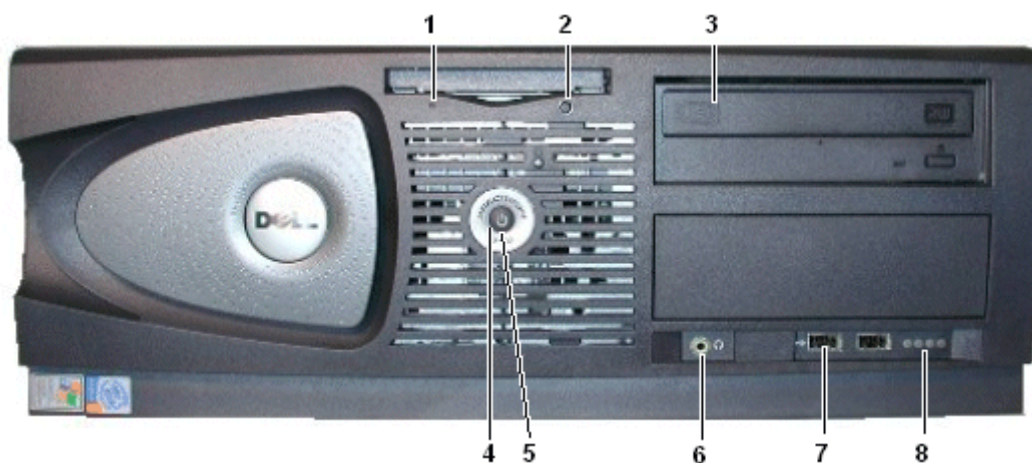


Figure 3-4 Front View of WS370 Enclosure

Table 3-2 WS370 Front Panel

No.	Control	Description
1	Floppy-drive light button	The floppy-drive light is on when the computer reads data from or writes data to the floppy drive. Wait until this light turns off before you remove the floppy disk from the drive.
2	Hard-drive light	The hard-drive light is on when the computer reads data from or writes data to the hard drive. The light might also be on when a device such as your CD player is operating.
3	CD/DVD drive	Access the CD or DVD drive.

Table 3-2 WS370 Front Panel

No.	Control	Description
4	Power light	The power light illuminates and blinks or remains solid to indicate the following different states: No light — The computer is in the off state. Steady green —The computer is in a normal operating state. Blinking green —The computer is in a power-saving state. To exit from a power-saving state, press the power button or use the keyboard or the mouse if it is configured as a wake device in the Windows Device Manager. For more information about sleep states and exiting from a power-saving state, see "Power Button" in the <i>Dell Precision™ Workstation 360 User's Guide</i>. See "Diagnostic Lights" in the <i>Dell Precision™ Workstation 370 User's Guide</i> for a description of light codes that can help you troubleshoot problems with your computer.
5	Power button	Press this button to turn on the computer.
6	Headphone connector	Attach headphones.
7	USB 2.0 connectors (2)	Use the front USB connectors for devices that you connect occasionally, such as joysticks or cameras. It is recommended that you use the back USB connectors for devices that typically remain connected, such as printers and keyboards.
8	Diagnostic lights (4)	Use the lights to help you troubleshoot a computer problem based on the diagnostic code. For more information, see "Diagnostic Lights" in the <i>Dell Precision™ Workstation 370 User's Guide</i> .

3. Computer Operation

3.3. WS370 Overview

Inside WS370 enclosure

The following picture shows the inside of the WS370 enclosure and identifies the areas within the enclosure.

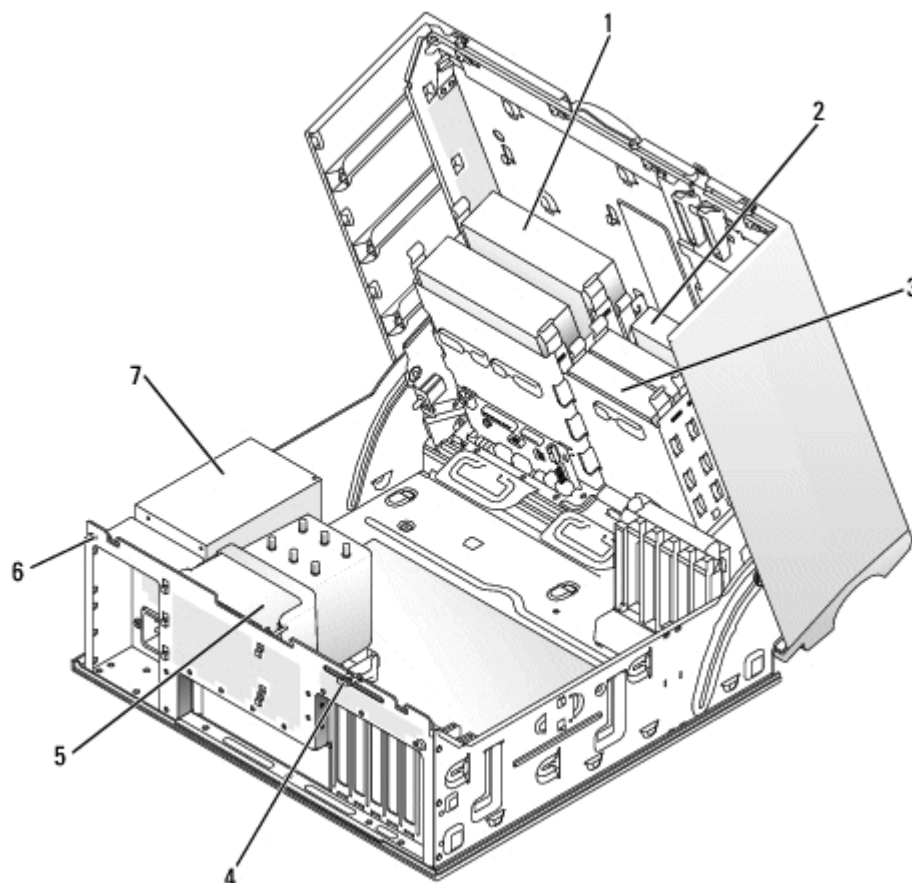


Figure 3-5 Inside WS370 Enclosure

- | | | | |
|---|--------------|---|---------------------------------------|
| 1 | CD/DVD drive | 5 | Microprocessor airflow shroud and fan |
| 2 | Floppy drive | 6 | Security cable slot |
| 3 | Hard drive | 7 | Power supply |
| 4 | Padlock ring | | |

Honeywell options for the WS370 (dual video)

The following picture shows the Honeywell options for a WS370 dual video configuration.

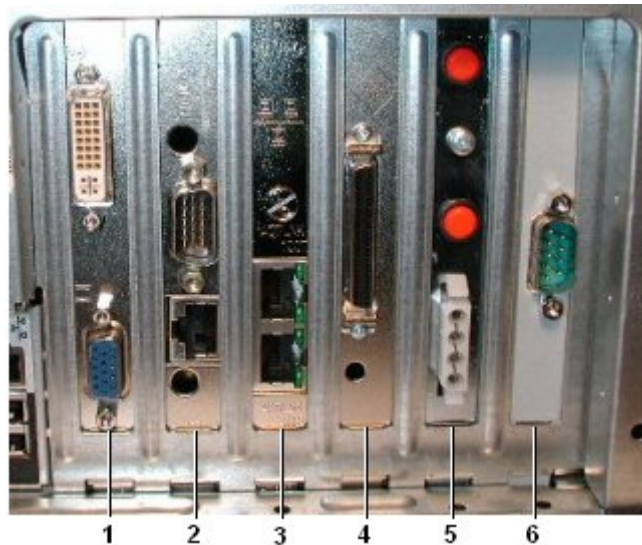


Figure 3-6 Honeywell Options for WS370 (dual video)

No.	Slot	Description of Card
1	PCI express	Dual video card (one or two video configuration)
2	PCI1	LCNP4 board
3	PCI2	Dual NIC (optional)
4	PCI3	SCSI controller card (optional)
5	PCI4	Remote power/circuit breaker assembly (optional)
6	PCI5	Serial port adapter (optional)

3. Computer Operation

3.3. WS370 Overview

Honeywell options for the WS370 (quad video)

The following picture shows the Honeywell options for a WS370 quad video configuration.

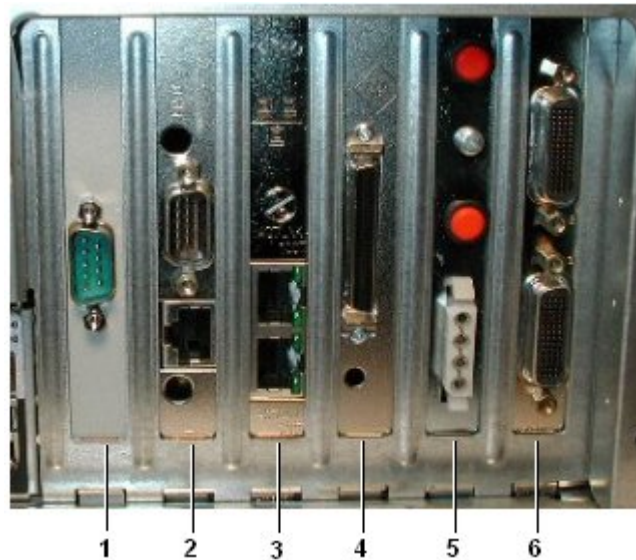


Figure 3-7 Honeywell Options for WS370 (quad video)

No.	Slot	Description of Card
1	PCI express	Serial port adapter (one or two video configuration)
2	PCI1	LCNP4 board
3	PCI2	Dual NIC (optional)
4	PCI3	SCSI controller card (optional)
5	PCIX1x	Remote power/circuit breaker assembly (optional)
6	PCI4	Quad video card (three or four video configuration)

Device cable Colors

The devices within the enclosure have specific colors as listed in the following table.

Device	Cable Color
Hard drive	Blue pull tab
Floppy drive	Black pull tab
CD/DVD drive	Orange pull tab
I/O panel	Yellow pull tab

3.4 WS470 Overview

Front view of WS470

The following figure shows the front view of the WS470. Refer to Table 3-3 for a complete description of the callouts.

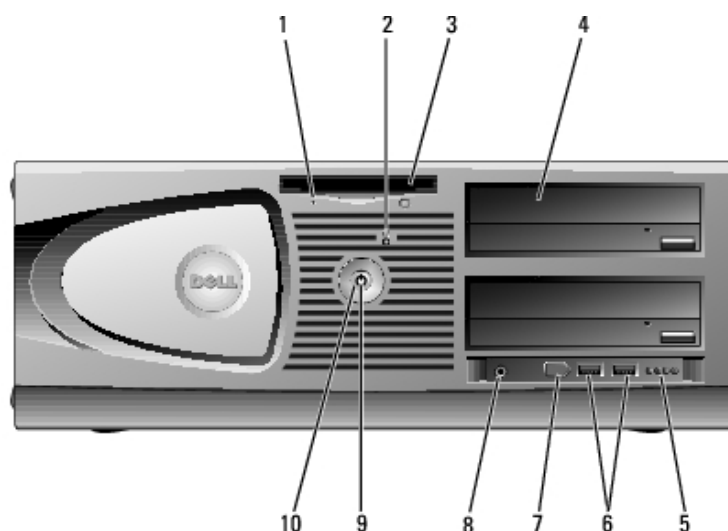


Figure 3-8 Front View of WS470 Enclosure

Table 3-3 WS470 front panel

No	Item	Description
1	floppy-drive activity light	The floppy drive light is on when the computer reads data from or writes data to the optional floppy drive. Wait until this light turns off before you remove the floppy disk from the drive.
2	hard-drive activity light	The hard drive light is on when the computer reads data from or writes data to the hard drive. The light might also be on when a device such as your CD player is operating.

Table 3-3 WS470 front panel

No	Item	Description
3	floppy drive (optional)	Access the optional floppy drive.
4	CD/DVD drive(s)	Access additional drives, such as a CD or DVD drive. Drive arrangement varies on computer configuration.
5	diagnostic lights (4)	Use the lights to help you troubleshoot a computer problem based on the diagnostic code. For more information, see "Diagnostic Lights."
6	USB 2.0 connectors (2)	Use the front USB connectors for devices that you connect occasionally, such as flash memory key or cameras, or for bootable USB devices (see "System Setup" for more information on booting to a USB device). It is recommended that you use the back USB connectors for devices that typically remain connected, such as printers and keyboards.
7	IEEE 1394a connector	Attach high-speed serial multimedia devices, such as digital video cameras. NOTE: This connector is available if you purchase an add-in IEEE 1394a card, "Firewire".
8	headphone connector	Use the headphone connector to attach headphones and most kinds of speakers.

3. Computer Operation

3.4. WS470 Overview

Table 3-3 WS470 front panel

No	Item	Description
9	power light	The power light illuminates and blinks or remains solid to indicate different states: No light — The computer is turned off. Steady green — The computer is in a normal operating state. Blinking green — The computer is in a power-saving state. Blinking or solid yellow— See "Power Problems." To exit from a power-saving state, press the power button or use the keyboard or the mouse if it is configured as a wake device in the Windows Device Manager. For more information about sleep states and how to exit from a power-saving state, see "Power Management." See "Diagnostic Lights" for a description of light codes that can help you troubleshoot problems with your computer.
10	power button	Press to turn on the computer. NOTICE: To avoid losing data, do not use the power button to turn off the computer. Instead, perform an operating system shutdown.

Rear view of WS470

The following figure shows the rear view of the WS470. Refer to Table 3-4 for a complete description of the callouts.

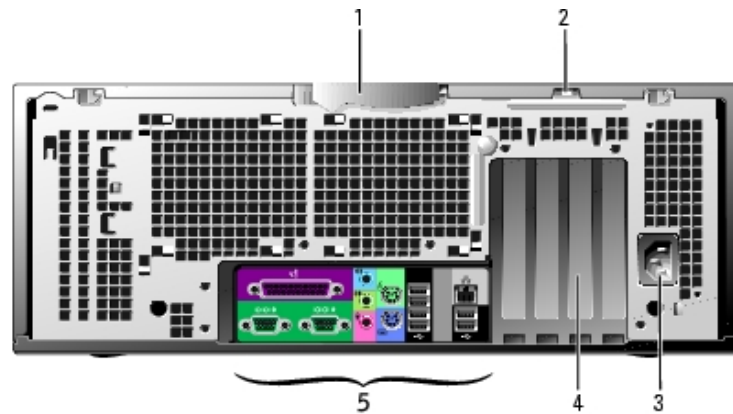


Figure 3-9 Rear View of WS470 Enclosure

Table 3-4 WS470 rear panel

No	Item	Description
1	cover latch release	To open the computer, lay the computer on its side with the cover latch release at the top and then push the latch to the left
2	padlock ring	Insert a padlock to lock the computer cover.
3	power connector	Insert the power cable.
4	card slots	Access connectors for any installed cards.
5	back panel connectors	Plug serial, USB, and other devices into the appropriate connectors.

Inside WS470 enclosure

The following picture shows the inside of the WS470 enclosure and identifies the areas within the enclosure.

3. Computer Operation

3.4. WS470 Overview

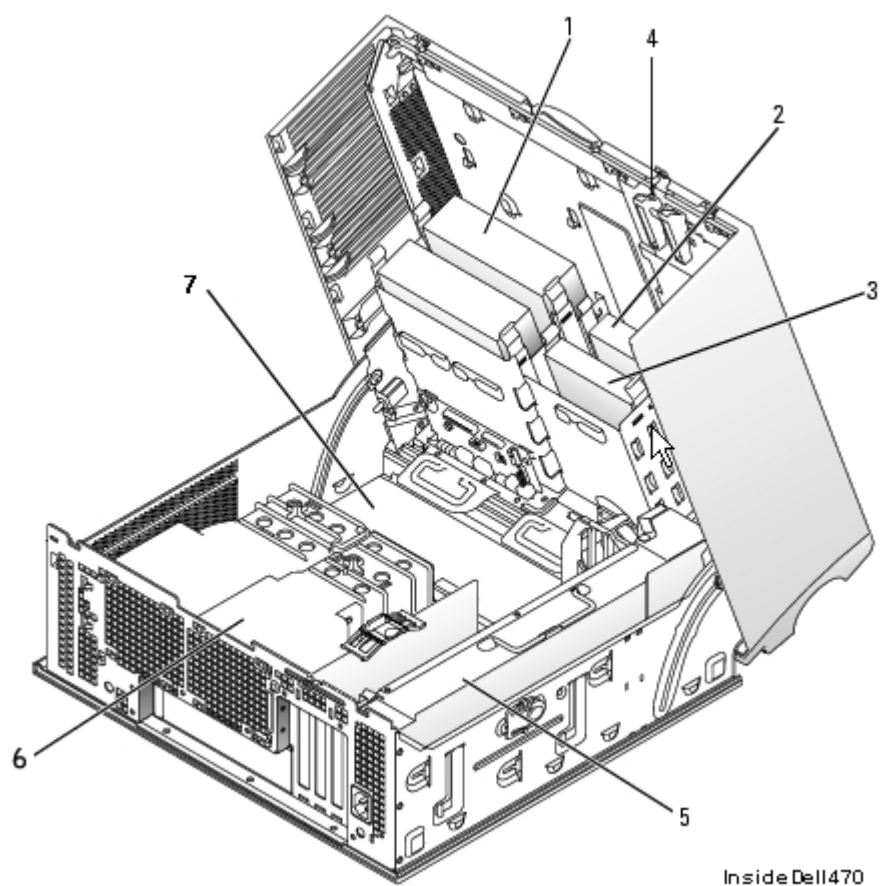


Figure 3-10 Inside WS470 Enclosure

- | | | | |
|---|--------------|---|---------------------------------------|
| 1 | CD/DVD drive | 5 | Power supply |
| 2 | Floppy drive | 6 | Microprocessor airflow shroud and fan |
| 3 | Hard drive | 7 | System board |
| 4 | Padlock ring | | |

Honeywell options for the WS470

The following picture shows the Honeywell options for a WS470 dual video configuration.

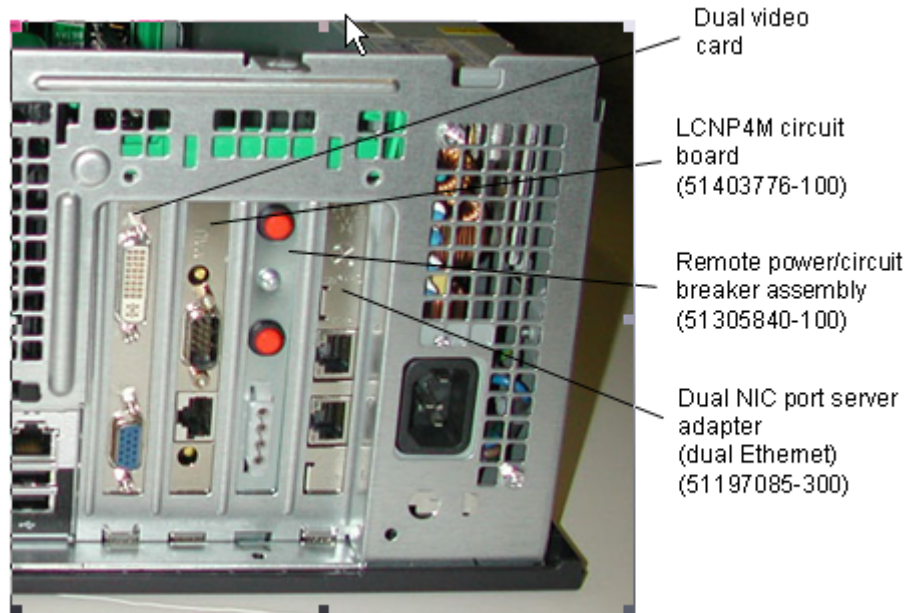


Figure 3-11 Honeywell Options for WS470

No.	Slot	Description of Card
1	PCIe x 16	Standard - Dual video card (ATI - FireGL) Optional - Quad video card (MATROX QID Quad)
2	PCI	LCNP4M board
3	PCIe x 4	Empty or power adapter harness
4	PCIX	If TPS System, optional SCSI controller card If FTE or Experion System, Single/Dual Intel Pro 1000 Ethernet

3. Computer Operation

3.4. WS470 Overview

Device cable Colors

The devices within the enclosure have specific colors as listed in the following table.

Device	Cable Color
Hard drive	Blue cable
Floppy drive	Black pull tab
CD/DVD drive	Orange pull tab
I/O panel	Yellow pull tab

3.5 Network Connections

Overview

Each workstation platform must be connected to an LCN network and/or an ETHERNET network.

Ethernet network

ETHERNET 10/100/1000 Base T connection is standard on the Dell WS360, WS370, and WS470 platforms. A Dual NIC option is available for FTE. The on board Ethernet connection must be disabled in the System BIOS prior to installation of the Dual Intel PRO 100/1000 controller. If you are using FTE, you must install the FTE cables according to the specific instructions in the *FTE Installation and Service Guide*.

LCN network

The connection to the LCN is made via an optional Local Control Network Processor (LCNP4M (mid size)) card. This card provides the communication path for the computer to other LCN modules. The LCNP4 consists of an LCNP4 Card, a MAU cable and the LCN MAU (Media Access Unit). The LCNP4 card is a full length in WS360 and WS370 PCI card that consumes one PCI slot. The LCNP4, MAU and MAU cable are required to connect to the LCN Network. For the WS470 the LCNP4M consists of a PCI/PCI –X Express 3.3 / 5.0 Volt LCNP4M medium size card, a MAU cable and the LCN MAU.

The LCN node address should be set to the address the customer requires. If the LCN address is not known then the node address should be set to zero (0). Setting the address to zero (0) allows the node to be connected to the LCN without the risk of an address conflict with some other node. This is consistent with the current LCN standard procedure.

Refer to *LCN System Installation*, Subsection 3.6 for the rules and techniques of installing an LCN cable system.

3. Computer Operation

3.5. Network Connections

LCN connections

The LCN Cable A and Cable B connections are made through a single cable from the LCNP4 board to the LCN Media Access Unit (MAU) contained in a metal

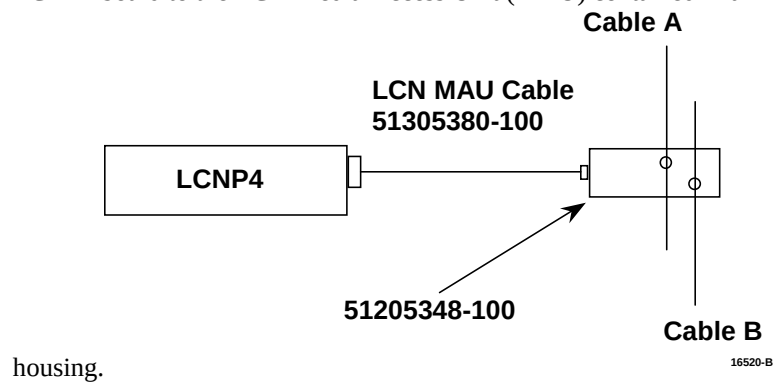


Figure 3-12 LCN MAU to LCN Cabling

MAU connection

Connect the MAU to both Cable A and Cable B coax T-connector as shown below.

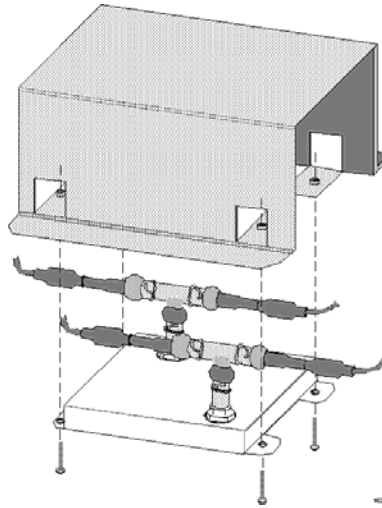


Figure 3-13 LCN MAU to LCN Cable T-Connections

3. Computer Operation

3.5. Network Connections

Z-Console connection to the LCN coax cable

The mounting for the LCN MAU is located in the footrest area of the station.

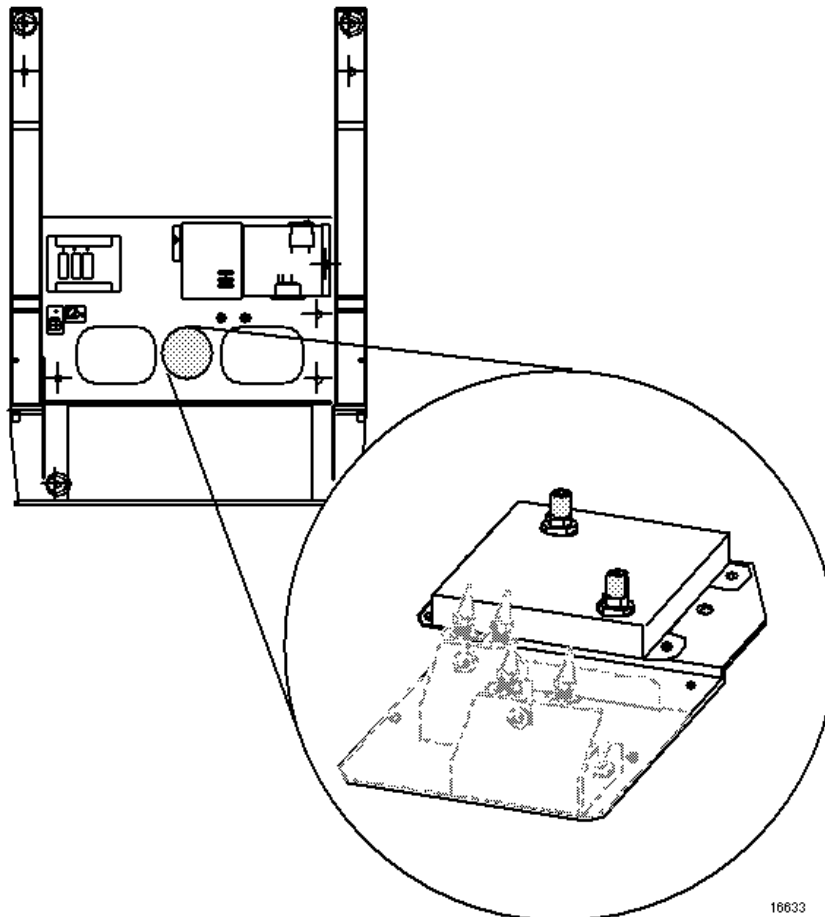


Figure 3-14 **Location of LCN MAU**

Connecting LCN coax to LCN MAU in Z-Console

Step	Action
1	Connect the color-coded A cables to the color-coded T-connector on the BNC connector marked A.
2	Connect the color-coded B cables to the color-coded T-connector on the BNC connector marked B.

Classic Console connection to the LCN Coax Cable

The LCN MAU is mounted to the EMC cover that is attached to the rear of the platform enclosure.

Step	Action
1	Follow the color-coding while connecting the color-coded A cables to the color-coded T-connector on the BNC connector marked A.
2	Follow the color-coding while connecting the color-coded B cables to the color-coded T-connector on the BNC connector marked B.

3. Computer Operation

3.5. Network Connections

4. Computer Servicing

4.1 Accessing the Enclosure

Overview

This section contains procedures for removing the computer from a Z console, Classic console and a cabinet. Although most procedures show the WS360, the procedures also apply to the WS370 and WS470.

Before you begin servicing



Attention

Perform a complete system shutdown before you begin any of the procedures in this section.



CAUTION

Before you begin any of the procedures in this section, follow the safety instructions in the *Dell System Information Guide*.



CAUTION

To avoid electrical shock, always unplug your computer from the electrical outlet before opening the cover.



Attention

Be careful when opening the computer cover to ensure that you do not accidentally disconnect cables from the system board.

4. Computer Servicing

4.1. Accessing the Enclosure

Removing the computer enclosure in a Z console (fixed mount)

If the computer is mounted using the fixed mount option, you will need to remove it from the furniture before servicing.

Step	Action
1	From the rear of the console, loosen the two screws on the two mounting bracket covers and remove the bracket covers.  A photograph showing the front of a black Dell computer enclosure. It features a large circular speaker on the left, a power button in the center, and two 5.25-inch floppy disk drives on the right. The enclosure is mounted within a light-colored metal console frame.
2	Slide the furniture bay cover off the mounting railings.  A photograph showing a large, dark grey plastic bay cover being slid off the front of the console. The cover is tilted downwards, revealing the internal components of the computer enclosure behind it. The console is mounted on a stand with red casters.

4. Computer Servicing
4.1. Accessing the Enclosure

Step	Action
3	Lift up slightly on the computer and slide it from the fixed mounting railings. 
4	Position the furniture bay cover back on the mounting railings and slide it all the way back.

4. Computer Servicing

4.1. Accessing the Enclosure

Removing the computer from a Classic console (fixed mount)

If the computer is mounted using the fixed mount option, you will need to remove it from the furniture before servicing.

Step	Action
1	Remove the console door to access the console enclosure.
2	If the mounting bracket covers are attached, loosen the two screws on each mounting bracket cover and remove them.
3	Slide the computer from the console mounting railings. 

Opening the enclosure

To simplify servicing when using the slide mount option, internal hardware components can be serviced without having to remove the enclosure from the Console or Cabinet.

Step	Action
1	If this is a slide mount computer, slide the tray from the furniture
2	From the back of the computer, slide the cover release latch to the left and gently lift up on the cover. 
3	Service the hardware components as required: • For servicing the LCNP4 board, see Section 4.2 in this document. • For servicing other Honeywell installed options see Section 4.3 in this document. • For other components, see the Dell service manual.

4.2 Servicing the LCNP4

LCNP4 board description

The LCNP4 board has 16 MB of on-board memory that is soldered to the board.

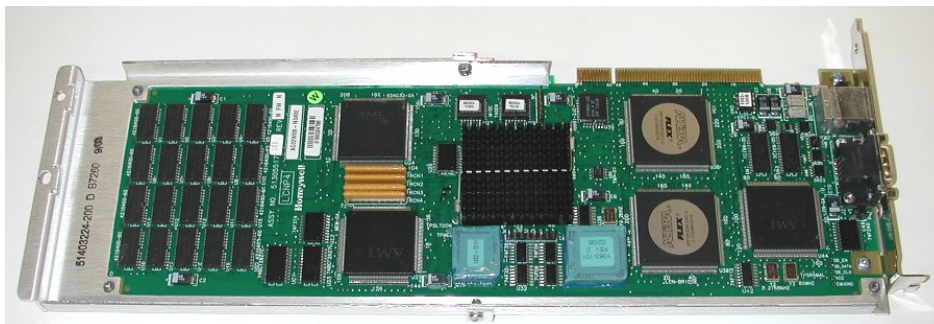


Figure 4-1 LCNP4 Board

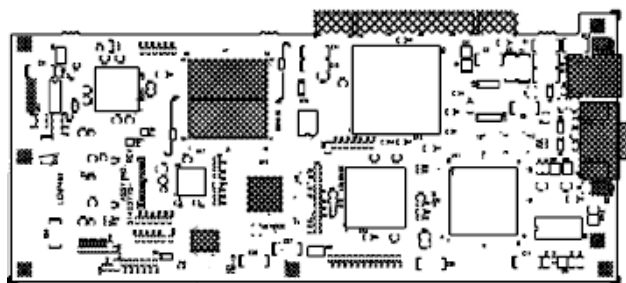


Figure 4-2 LCNP4M Board

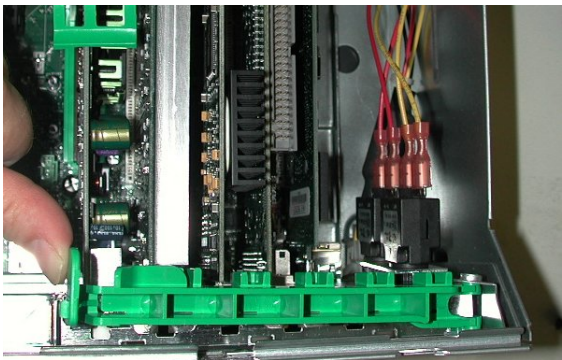
Replace the LCNP4 board

The LCNP4 board is located in the PCI1 slot for both the WS360 and WS370. The LCNP4M board is located in the PCI, slot 2 in the WS470. Use this procedure to replace the LCNP4 assembly.



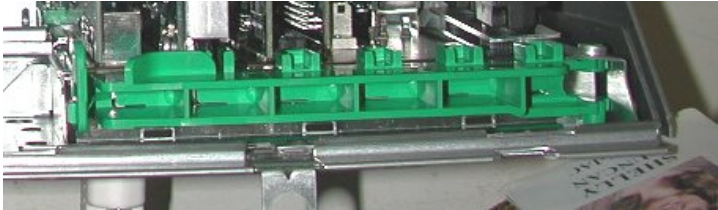
ESD HAZARD

The LCNP4 board is an electrostatically sensitive device. Use a grounding strap and grounded work surfaces and equipment when handling this component. Store and transport parts only in electrostatically safe containers.

Step	Action
1	Perform all necessary procedures in Section 4.1, "Accessing the Enclosure" to access the LCNP4 assembly including: • Shutting down system • Disconnecting power from the computer • Accessing the inside of the electronics enclosure
2	Disconnect the LCN MAU cable from the LCNP4 board.
3	Release the tab on the card retention arm, and lift it up to access the LCNP4 board. 

4. Computer Servicing

4.2. Servicing the LCNP4

Step	Action
4	While wearing a grounded ESD wrist strap, remove the LCNP4 board assembly from the PCI slot. Note that the pictures in this procedure show the LCNP4 (full size) board. The LCNP4M (medium or ½ size) board is used WS470. 
5	Verify that the new LCNP4 board has the correct LCN pin assignments.
6	Insert the new LCNP4 board assembly in the PCI1 slot. The WS470 uses PCI, slot 2 for the LCNP4M board assembly.
7	Replace the card retention arm making sure the tab locks into place. 
8	Replace the computer in the furniture using the appropriate installation procedures in Section 2, "Platform Installation".
9	Press the Power On/Off button on the front of the platform to turn the power back on.
10	Run the Configurator to re-establish the correct node number.
11	Reconnect the LCN MAU cable.
12	Restart the Operating System.

4.3 Replacing/Adding Other Options

Overview

This section contains procedures for adding or replacing optional components in the WS360, WS370 and WS470.

Before you begin



CAUTION

Be careful not to damage the EMI gasket fingers when removing/installing boards.



ESD HAZARD

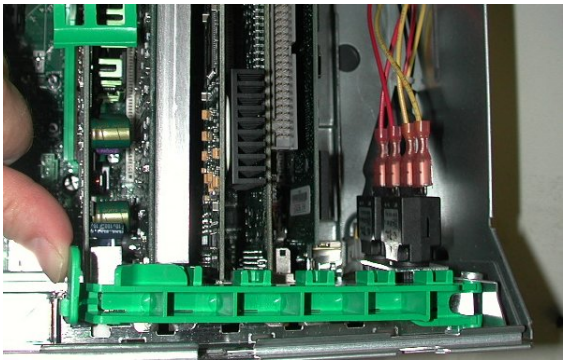
Use a grounding strap and grounded work surfaces and equipment when handling any electrostatically sensitive components such as the video cards, NIC adapter cards, and SCSI controller cards. Store and transport parts only in electrostatically safe containers.

4. Computer Servicing

4.3. Replacing/Adding Other Options

Replace/add video card

Use this procedure to replace a video card in the computer.

Step	Action
1	Perform all necessary procedures in Section 4.1, "Accessing the Enclosure" including: • Shutting down system • Disconnecting power from the computer • Accessing the inside of the electronics enclosure • Disconnecting cables from the card being replaced.
2	Replace the tab on the card retention arm and lift it up to access the video card slot. 
3	If necessary, remove the DVI to HG15 adapter from the video connector port on the video card.

4	If replacing the video card in the WS360 AGP slot, remove the video card retainer clip: • Push down gently on the retainer clip tabs • Lift up on the retainer clip to remove it. 		
5	While wearing a grounded ESD wrist strap, remove the existing video card from the slot.		
6	Insert the new video card in the proper slot: <table border="0" style="width: 100%;"> <tr> <td style="vertical-align: top; width: 50%;"> <u>WS360</u> Dual video card: AGP slot Quad video: PCI2 slot <u>WS370</u> Dual video card: PCI express slot Quad video: PCI4 slot </td><td style="vertical-align: top; width: 50%;"> <u>WS470</u> Single/Dual video card: PCI express, slot 1 Quad video card: PCIX, slot 1 Note: Do not mix flat panel displays and CRTs in a multi-screen configuration on a single platform. </td></tr> </table>	<u>WS360</u> Dual video card: AGP slot Quad video: PCI2 slot <u>WS370</u> Dual video card: PCI express slot Quad video: PCI4 slot	<u>WS470</u> Single/Dual video card: PCI express, slot 1 Quad video card: PCIX, slot 1 Note: Do not mix flat panel displays and CRTs in a multi-screen configuration on a single platform.
<u>WS360</u> Dual video card: AGP slot Quad video: PCI2 slot <u>WS370</u> Dual video card: PCI express slot Quad video: PCI4 slot	<u>WS470</u> Single/Dual video card: PCI express, slot 1 Quad video card: PCIX, slot 1 Note: Do not mix flat panel displays and CRTs in a multi-screen configuration on a single platform.		
7	If you replaced the video card in the AGP slot, replace the video card retainer clip making sure it rests securely against the top of the video card as shown: 		
8	Replace the card retention arm making sure the tab locks into place. 		

4. Computer Servicing

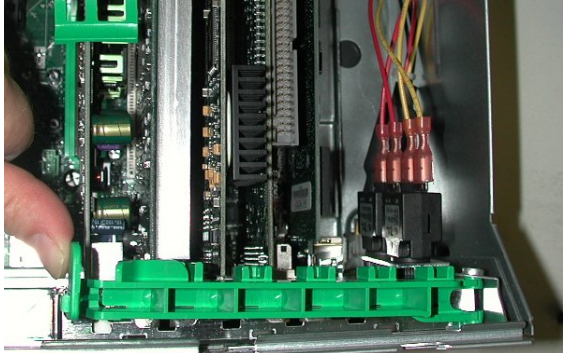
4.3. Replacing/Adding Other Options

9	Refer to one of the following when reconnecting the video displays: <table><tr><td data-bbox="475 474 873 527"><u>WS360</u>: Table 2-5, WS360 Video Channels</td><td data-bbox="894 474 1281 558"><u>WS470</u>: Connect remaining WS470 cables and power in section 2.8 Connecting Cables to a WS470.</td></tr><tr><td data-bbox="475 533 873 588"><u>WS370</u>: Table 2-6, WS370 Connectors (quad video)</td><td></td></tr><tr><td data-bbox="475 594 873 644">Table 2-7, WS370 Connectors (dual video)</td><td data-bbox="894 594 1281 644">For an Icon system: Connect Icon monitors</td></tr></table>	<u>WS360</u> : Table 2-5, WS360 Video Channels	<u>WS470</u> : Connect remaining WS470 cables and power in section 2.8 Connecting Cables to a WS470.	<u>WS370</u> : Table 2-6, WS370 Connectors (quad video)		Table 2-7, WS370 Connectors (dual video)	For an Icon system: Connect Icon monitors
<u>WS360</u> : Table 2-5, WS360 Video Channels	<u>WS470</u> : Connect remaining WS470 cables and power in section 2.8 Connecting Cables to a WS470.						
<u>WS370</u> : Table 2-6, WS370 Connectors (quad video)							
Table 2-7, WS370 Connectors (dual video)	For an Icon system: Connect Icon monitors						

Replace/add SCSI controller

Use this procedure to replace or add a SCSI controller card in PCI slot 3 on WS360, or on WS470; slot 4 (the optional WS470 SCSI interface).

Note that the WS470 uses a USB/SCSI adapter so you can use a USB port as the interface for a SCSI device.

Step	Action
1	Perform all necessary procedures in Section 4.1, "Accessing the Enclosure" including: • Shutting down system • Disconnecting power from the computer • Accessing the inside of the electronics enclosure Disconnecting cables to the card being replaced.
2	Release the tab on the card retention arm, and lift it up to access the SCSI controller card slot. 
3	If you are replacing the existing SCSI controller, remove it from the slot.
4	Insert the new SCSI controller card in the PCI3 slot on WS360 and WS370; PCIX, slot 4 on WS470.

4. Computer Servicing

4.3. Replacing/Adding Other Options

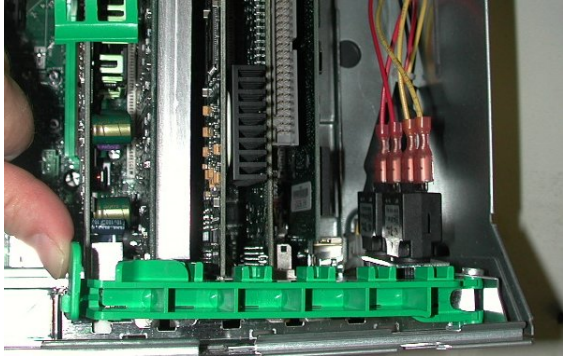
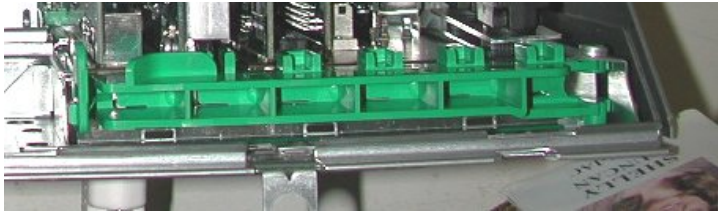
5

Replace the card retention arm and make sure the tab locks into place.



Replace/add the Dual NIC

Use this procedure to replace or add the dual NIC card. If you are adding the FTE dual NIC, you must have access to the *FTE Installation and Service Guide*.

Step	Action
1	Perform all necessary procedures in Section 4.1, "Accessing the Enclosure" to access the video card in the PCI slots, including: • Shutting down system • Disconnecting power from the computer • Accessing the inside of the electronics enclosure Disconnecting cables to the card being replaced.
2	Release the tab on the card retention arm, and lift it up to access the NIC card slot. 
3	If you are replacing an existing Dual NIC, remove it from the slot.
4	Insert the new dual NIC card in the proper slot: WS360: PCI4 slot WS470: PCIX, slot 4 WS370: PCI2 slot
5	Replace the card retention arm making sure the tab locks into place. 

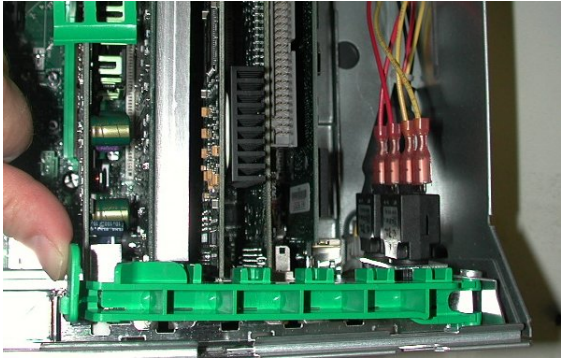
4. Computer Servicing

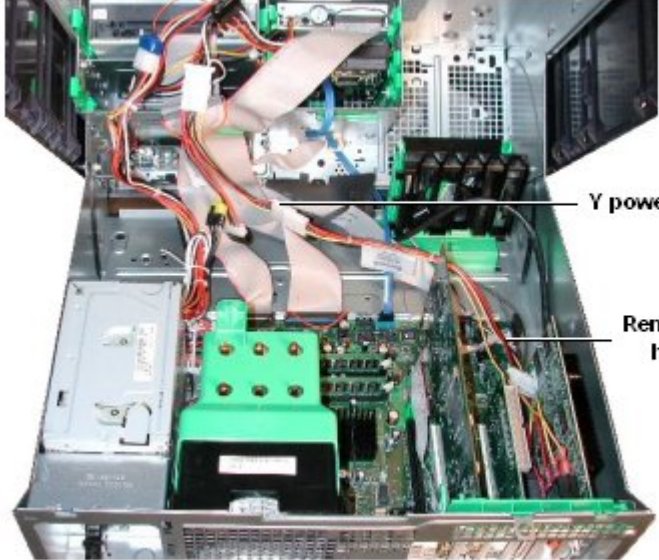
4.3. Replacing/Adding Other Options

6	You will need to reconnect your FTE cables according to the procedures in the <i>FTE Installation and Service Guide</i> in order to verify that the two connectors on the Dual NIC are connected to the right switches.
---	---

Replace/add the remote power/circuit breaker assembly

Use this procedure to replace or add the remote power/circuit breaker assembly.

Step	Action
1	Perform all necessary procedures in Section 4.1, "Accessing the Enclosure" to access the video card in the PCI slots, including: • Shutting down system • Disconnecting power from the computer • Accessing the inside of the electronics enclosure • Disconnecting cables to the card being replaced.
2	Release the tab on the card retention arm, and lift it up to access the remote power card slot. 
3	Disconnect the 4-pin Molex connector from the Y cable and remove the Remote power/circuit breaker assembly.
4	Insert the new Remote power/circuit breaker assembly in the proper slot: WS360: PCI4 slot WS370: PCI2 slot WS470: PCIe x 4, slot 3

Step	Action
5	Connect the 4-pin Molex connector on the remote power/circuit breaker assembly's cable harness to the Y power cable: 

4. Computer Servicing

4.3. Replacing/Adding Other Options

Add additional memory to the WS370 or WS470



ESD HAZARD

Use a grounding strap and grounded work surfaces and equipment when handling memory

Use this procedure to add two additional gigabits of memory to the WS370 or WS470.

Step	Action
1	Perform all necessary procedures in Section 4.1, "Accessing the Enclosure" to access the memory slots, including: • Shutting down system • Disconnecting power from the computer • Accessing the inside of the electronics enclosure
2	Place the memory in the slots according to the configuration in: WS370 Figure 1-6, WS370 3 GB Memory Configuration. WS470 Figure 1-9 WS470 3 GB Memory Configuration
3	Verify the memory modules are seated properly.
4	Close the enclosure and reconnect the power cable.
5	Restart the computer.

4.4 Spare Parts Lists

Ordering spare parts

This section contains lists of spare parts available from Honeywell. Additional parts can be ordered directly from Dell by calling 1-800-456-3355. Before calling, have your service tag number available.

WS360 enclosure spare parts from Honeywell

The following table lists the spare parts available from Honeywell for the WS360 enclosure.

Table 4-1 Spare Parts for WS360 Enclosure

Description	Part No.
LCNP4 Assembly	51403299-100
Dual NIC for FTE	51197085-200
Remote Power/Circuit Breaker Assembly	51305840-100
Expansion RAM 1GB, 72-bit, 400MHZ, DDR, ECC, SDRAM, as two 512 MB DIMMS	51199585-903

WS370 enclosure spare parts from Honeywell

The following table lists the spare parts available from Honeywell for the WS370 enclosure.

Table 4-2 Spare Parts for WS370 Enclosure

Description	Part No.
LCNP4 Assembly	51403299-100
Dual NIC for FTE	51197085-200
Remote Power/Circuit Breaker Assembly	51305840-100
Expansion RAM 1GB, 533 MHZ, DDR2, ECC, SDRAM, as two units of 512 MB DIMMs	51199586-903

4. Computer Servicing

4.4. Spare Parts Lists

Other WS360/WS370/WS470 spare parts from Honeywell

The following table lists part numbers for other items (such as a console mount for the platform) that may be used with the WS360, WS370 and the WS470.

Table 4-3 Spare Parts for WS360/WS370/WS470 Enclosure

Description	Part No.
Blank panel front cover, 1.75-inch height	51201248-100
Air duct baffle, 1.75-inch height	51303521-100
Blank panel front cover, 3.5-inch height	51201248-200
Air duct baffle, 3.5-inch height	51303521-200
Blank panel front cover, 5.25-inch height	51201248-300
Air duct baffle, 5.25-inch height	51303521-300
Blank panel front cover, 7-inch height	51201248-400
Air duct baffle, 7-inch height	51303521-400
Screw, M5 x 16, self tapping for blank panel front covers	51195168-616
Screw, M5 x 12, hex for air baffles	51108888-612
Washer, cup/ground (8 for front covers, 4 for air baffles)	51109931-100

WS470 enclosure spare parts from Honeywell

The following table lists the spare parts available from Honeywell for the WS470 enclosure.

Table 4-4 Spare Parts for WS470 Enclosure

Description	Part No.
LCNP4M Assembly	51403776-100
Dual NIC for FTE	51197085-200
Remote Power/Circuit Breaker Assembly	51305840-100

4.5 Troubleshooting

System interrupt assignments for WS360 with Carroll Touchscreen

These PCI Bus devices are dynamically reconfigurable by the system. The actual IRQs assigned may change from one boot to the next. They may even show overlapping IRQ assignments among themselves. This is normal.

IRQ	Device	Notes
1	Standard 101/102-key or Microsoft Natural PS/2 Keyboard	PC Keyboard Port
4	Communication Port (COM3)	Serial port 1 on motherboard COM 1
3	Communication Port (COM4)	Serial port 2 on motherboard COM 2
5	Honeywell LCNP4 GPS	LCNP4
18	Intel ® 82540EM Base Network Connection	Ethernet
06	Standard Floppy disk controller	Floppy
16	NVIDIA Quadro FX500	AGP
22	Matrox G450 Multimonitor	PCI
22	Matrox G450 Multimonitor	PCI
18	Symbios Logic 875xs/ Dm 2280x PCI SCSI Adapter	LSI SCSI Controller
17	SondMAX Integrated Digital Audio	On board sound
12	Microsoft PS/2 Mouse	Mouse Port

4. Computer Servicing

4.5. Troubleshooting

System interrupt assignments for WS370 with Carroll Touchscreen

IRQ	Device	Notes
1	Standard 101/102-key or Microsoft Natural PS/2 Keyboard	PC Keyboard Port
4	Communication Port (COM3)	Serial port 1 on motherboard COM 1
3	Communication Port (COM4)	Serial port 2 on motherboard COM 2
10	Honeywell LCNP4GPS	LCNP4
18	Intel ® PRO/100 Base Network Connection	Ethernet
06	Standard Floppy disk controller	Floppy
16	ATI FireGL V3100	PEG
10	Intel ® 82801FB/FBM SMBus Controller	Ethernet
23	SondMAX Integrated Digital Audio	On board sound
12	Microsoft PS/2 Compatible Mouse	Mouse Port

System interrupt assignments for WS470 with Carroll Touchscreen

IRQ	Device	Notes
1	Standard 101/102-key or Microsoft Natural PS/2 Keyboard	PC Keyboard Port
4	Communication Port (COM3)	Serial port 1 on motherboard COM 1
3	Communication Port (COM4)	Serial port 2 on motherboard COM 2
6	Standard Floppy disk controller	Floppy
10	Honeywell LCNP4GPS	LCNP4
10	Intel ® 82801FB/FBM SMBus Controller	Ethernet
12	Microsoft PS/2 Compatible Mouse	Mouse Port
16	ATI FireGL V3100 / Matrox QID LP	PEG / PCIe
17	SondMAX Integrated Digital Audio	On board sound
49	Intel Pro / 1000 Desktop adapter	Ethernet

Ethernet Troubleshooting

There are two indicators located in the RJ-45 connector on the back of the NIC card in the platform.

The left indicator is the Link Integrity Indicator and the right indicator is the Activity Indicator.

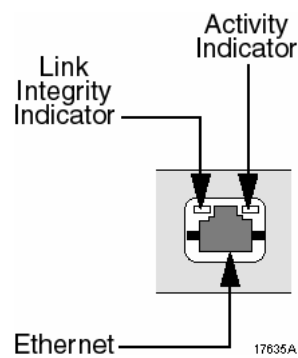


Figure 4-3 Ethernet Indicators on the RJ-45 Connector

4.6 Verifying Correct BIOS Settings

Purpose

Honeywell configures specific BIOS settings in the factory for each of the computer platform configurations, and these settings should not be altered. BIOS settings for the WS370 are listed in Table 4-5 and BIOS settings for the WS470 are listed in Table 4-6 so that you may verify the correct settings.

Accessing BIOS

Use this procedure to access BIOS and view the settings. **DO NOT** attempt this procedure unless you are familiar with BIOS.

Step	Action
1	Restart the computer
2	When the power on self-test screens appear, press F2 to enter the BIOS Setup.

WS370 BIOS Settings

This table lists the BIOS settings configured in the factory for the WS370 computer.

Table 4-5 WS370 BIOS Settings

System Info	
System	Dell System Workstation 370
BIOS Version	A02
Processor Info	
Type	Intel P4 CPU 3.40 GHz
Clock Speed	3.40GHz
Bus Speed	800MHz
L2 Cache	1MB
64 Bit	Yes
Memory Info	
Speed	533 MHz
Technology	DDR2 SDRAM
Boot Sequence	1. Onboard or USB CD-ROM Drive 2. Onboard or USB Floppy Drive 3. Onboard SATA Hard Drive 4. Onboard IDE Hard Drive (not present)

4. Computer Servicing

4.6. Verifying Correct BIOS Settings

Table 4-5 WS370 BIOS Settings

Drives	
Diskette Drive	Internal
Drive 0: SATA 0	ON
Drive 1: SATA 1	OFF
Drive 2: SATA 2	OFF
Drive 3: SATA 3	OFF
Drive 4: PATA 0	ON
Drive 5: PATA 1	OFF
SATA Operation	RAID Autodetect / AHCI
Smart Reporting	ON
Onboard Devices	
Integrated NIC	ON OFF if FTE is installed
Integrated Audio	ON
USB Controller	No-Boot
LPT Port Mode	PS/2
LPT Port Address	378H
Serial Port 1	AUTO
Serial Port 2	AUTO
PS/2 Mouse Port	ON
Video	
Primary Video	PEG (ATI V3100, default) PCI (Matrox G450)
Performance	
Hyper-Threading	OFF (W2K, default), ON (Windows XP)
HDD Acoustic Mode	Suggested
Security	
Password Changes	Unlocked
Chassi Intrusion	On-Silent
Intrusion Alert	Acknowledge
Power Management	
AC Recovery	ON
Auto Power On	OFF
Low Power Mode	OFF
Remote Wakeup	OFF
Suspend Mode	S1

Table 4-5 WS370 BIOS Settings

Maintenance	
Load Defaults	Cancel
Event Log	Mark All Entries
POST Behavior	
Fast Boot	OFF
Numlock Key	ON
Post Hotkeys	Setup & Boot Menu
Keyboard Errors	Report

WS470 BIOS Settings for Dual Processor

This table lists the BIOS settings configured in the factory for the WS470 dual processor computer.

Table 4-6 WS470 BIOS Settings for Dual Processor

System Info	
System	Dell System Workstation 470
BIOS Version	A03 (02/02/2005) or greater
Processor Info CPI 0	
Type	Intel® XEON™ CPU 3.40 GHz or greater
Clock Speed	3.40GHz or greater
Bus Speed	800MHz or greater
L2 Cache	2 MB or greater
ID	0F41 or equivalent
64 Bit	Yes
Processor Info CPI 1	
Type	Intel® XEON™ CPU 3.40 GHz or greater
Clock Speed	3.40GHz or greater
Bus Speed	800MHz or greater
L2 Cache	2 MB or greater
ID	0F41 or equivalent
64 Bit	Yes

4. Computer Servicing

4.6. Verifying Correct BIOS Settings

Table 4-6 WS470 BIOS Settings for Dual Processor

Drive 0: SATA 0	ON
Drive 1: SATA 1	OFF
Drive 2: SATA 2	OFF
Drive 3: SATA 3	OFF
Drive 4: PATA 0	ON
Drive 5: PATA 1	OFF
SATA Operation	RAID Off
Smart Reporting	Off
Onboard Devices	
Integrated NIC	ON OFF if FTE is installed for TPS systems
Integrated Audio	ON
USB Controller	On
LPT Port Mode	PS/2
LPT Port Address	378H
Serial Port 1	AUTO
Serial Port 2	AUTO
PS/2 Mouse Port	ON
Video	
Primary Video	PEG (ATI V3100, default) PCIe x 16 (Matrox QID)
Performance	
CPU Count	ON
Hyper-Threading	OFF
HDD Acoustic Mode	Bypass
Security	
Admin Password	Not set
System Password	Not set
Password Changes	Unlocked
Chassi Intrusion	On-Silent
Intrusion Alert	Acknowledge
Execute Disable	ON
Power Management	
AC Recovery	OFF
Auto Power Time	12:00 AM
Auto Power On	OFF
Low Power Mode	OFF

Table 4-6 WS470 BIOS Settings for Dual Processor

Remote Wakeup	OFF
Suspend Mode	S3
Maintenance	
Load Defaults	Cancel
Event Log	Mark All Entries
POST Behavior	
Fast Boot	OFF
Numlock Key	ON
Post Hotkeys	Setup & Boot Menu
Keyboard Errors	Report

WS470 BIOS Settings for Single Processor

This table lists the BIOS settings configured in the factory for the WS470 single processor computer.

Table 4-7 WS470 BIOS Settings for Single Processor

System Info	
System	Dell System Workstation 470
BIOS Version	A03 (02/02/2005) or greater
Processor Info	
Type	Intel® XEON™ CPU 3.40 GHz or greater
Clock Speed	3.40GHz or greater
Bus Speed	800MHz or greater
L2 Cache	1 MB or greater
ID	0F41 or equivalent
64 Bit	Yes

4. Computer Servicing

4.6. Verifying Correct BIOS Settings

Table 4-7 WS470 BIOS Settings for Single Processor

Drives	
Diskette Drive	Internal
Drive 0: SATA 0	ON
Drive 1: SATA 1	OFF
Drive 2: SATA 2	OFF
Drive 3: SATA 3	OFF
Drive 4: PATA 0	ON
Drive 5: PATA 1	OFF
SATA Operation	RAID Off
Smart Reporting	Off
Onboard Devices	
Integrated NIC	ON OFF if FTE is installed for TPS systems
Integrated Audio	ON
USB Controller	On
LPT Port Mode	PS/2
LPT Port Address	378H
Serial Port 1	AUTO
Serial Port 2	AUTO
PS/2 Mouse Port	ON
Video	
Primary Video	PEG (ATI V3100, default) PCI (Matrox QID)
Performance	
CPU Count	ON
Hyper-Threading	OFF
HDD Acoustic Mode	Bypass
Security	
Admin Password	Not set
System Password	Not set
Password Changes	Unlocked
Chassi Intrusion	On-Silent
Intrusion Alert	Acknowledge
Execute Disable	ON
Power Management	
AC Recovery	OFF

Table 4-7 WS470 BIOS Settings for Single Processor

Auto Power On	OFF
Auto Power Time	12:00 AM
Low Power Mode	OFF
Remote Wakeup	OFF
Suspend Mode	S3
Maintenance	
Load Defaults	Cancel
Event Log	Mark All Entries
POST Behavior	
Fast Boot	OFF
Numlock Key	ON
Post Hotkeys	Setup & Boot Menu
Keyboard Errors	Report

Honeywell

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